

THE RADIUS SPECTRUM OF GAMES, AND SIMULTANEOUS *Go* VIA OBJECTIVE QUANTUM REDUCTION

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ABSTRACT. Based on the theory of simultaneization developed here, we introduce an invariant of combinatorial games, formalized as automata, valued in subsets of $\mathbb{N} \sqcup \{\infty\}$ and called the radius spectrum. It is an invariant of bisimulation: it reads the dynamics alone, up to behavioral equivalence, and is orthogonal to quantitative data such as cardinality, and to the scoring. We then draw a stark line between chess and *Go*: the radius spectrum of *Go* is all of $\mathbb{N} \sqcup \{\infty\}$, of chess $\{0\}$. The proof factors through the construction of a particular simple symmetric simultaneization *SGo* of *Go*, all terms being mathematical notions. *SGo* is striking purely as a game: it is refereed by the display alone, and it is sharper and unreasonably deeper (three times in our particular test) than *Go* in computer simulations. Its rules are simple, especially mathematically; in fact the Komi, Ko and suicide rules of *Go* are removed. For these reasons we take time to describe it gently and fully. The paper is formalized in Lean, but human readable proofs are given here for everything. The construction of *SGo* is inspired by physical quantum theory, despite the game being combinatorial.

1. INTRODUCTION

At a basic level we study simultaneizations of combinatorial games formalized as automata and use this to derive qualitative invariants of their game dynamics. There are intrinsic reasons why one may want to simultaneize a game – in particular *Go*. The latter is an ancient game governed by local-to-global geometric principles. It is a sequential game (that is, turns alternate between the two players), which has the benefit of a very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. But for *Go* this structure creates asymmetries which force departures from local-to-global principles – mainly Ko fights.¹ One idea is that with simultaneization we can remove the Ko rule completely, and of course Komi. A more general reason to study simultaneizations comes from the concurrency problems; we sketch it at the end of this introduction.

The basic finding is that perhaps due to its local-to-global nature, *Go* is almost simultaneous. We make this precise by proving that *Go*, on a board of side at least five, has simple symmetric simultaneizations of faithfulness radius any $k \in \mathbb{N}$ or ∞ , see Theorem 6.3. Much of the general theory of simultaneizations is developed in the present paper.

The faithfulness radius calculations lead us to introduce a bisimulation invariant of the dynamics of a game G , orthogonal to quantitative data such as cardinality, and to the scoring – the radius spectrum of G , valued in subsets of $\mathbb{N} \sqcup \{\infty\}$. For *Go* it is $\mathbb{N} \sqcup \{\infty\}$, by Theorem 6.3; in contrast, for chess, augmented with a pass move

¹If a player engages in a Ko fight, they must make moves that are strategically disconnected.

(Example 4.4), it is $\{0\}$, by Theorems 6.7 and 6.9. The paper has been formalized in Lean².

1.1. **SGo.** The proof of Theorem 6.3 factors through the construction of the radius zero game *SGo*, and it is the star here – it is an easy to play game with razor sharp and deep dynamics. It is a symmetric simultaneous combinatorial game, which is played on a standard *Go* board and refereed by the display alone³, although we need a third type of stone that we call q-stones, represented by colored stones. It is in some ways a simpler game than *Go* as it loses the Komi, Ko and suicide rules, and it has just two rules: stone placement and capture (aside from the end of game rule). Like classical *Go*, *SGo* has very robust game dynamics. We make this precise via some empirical studies outlined in Section 2.1 and Section 7.2. In computer simulations random games of *SGo* appear to be sharper than those of *Go*, markedly so at 19×19 (but this must be properly interpreted). More interestingly, *SGo* is a much deeper game than *Go* on 5×5 , with an Elo ladder three times the height. These statistics are preliminary, but *SGo* being deeper should be unsurprising from the construction alone, as the state space is much larger and the evolution rule more intricate.

The idea for *SGo* is loosely inspired by quantum mechanics. The rough idea is to branch the game state whenever the moves of the two players do not commute or otherwise interfere in classical *Go*. This is part of a more general theory of universal simultaneization of certain sequential games developed in Section 4.2, and this may be of independent interest. However, the universal simultaneization generally has uninteresting dynamics; in the case of *Go* this is seen empirically, cf. Section 2.1. To get interesting dynamics, the *SGo* game evolution (capture rule) invokes “objective reduction of the quantum state” – it is a deterministic, generally easy, objective operation performed on the board. The term objective reduction was coined by Penrose, see for instance [26]; in terms of the branch semantics we are collapsing certain branches.

Of course, the word quantum is used figuratively: there is no unitary time evolution, no complex amplitudes, and no probabilistic outcomes. The game is combinatorial and deterministic, and it would not be appropriate to call it “quantum *Go*”. For games with genuine quantum features see for instance [7], and for a quantum version of *Go* itself [29].

Using the Nash equilibrium theorem [24], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said, finding this even approximately is likely only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw, not just to draw on average; on the 2×2 board both players do, but already on the 3×3 board neither player has one, so optimal play is necessarily mixed (Section 5.1).

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [2], [31]. It also led to Conway’s theory of surreal numbers [4]. There is also research on simultaneous and synchronized combinatorial games, see for instance [3], [13], [14]. We expect that *SGo* will provide a simple but rich example that can help with further study of such games.

²<https://yashamon.github.io/web2/papers/sgoleanformalization/>

³Note that this is beyond even chess and *Go*: the latter have, for instance, castling rights and Ko rules that are not purely display driven.

1.2. Generalizations. The simultaneization theory generalizes to N players. The turn grading takes values in $\mathbb{Z}_N$, and the serialization map, defined when each single move is, collects the maximal defined prefixes of the $N!$ orders of play; for $N = 2$ this is the map P of Section 4.2. Games of simultaneous moves among N players are the normal form of non-cooperative game theory [40, 24]; their repeated play on a state space is a stochastic game [34, 9], in computer science a concurrent game structure [1], see also [35]. What is added here is the passage from a sequential game to a simultaneous one, and the measure of its fidelity.

The objective reduction is not special to *SGo* either. Its pattern is general: keep every serialization of the concurrent actions as a branch, and let objective information of the shared state prune the branches. This suggests an approach to concurrency. When the actions of N actors carry no reliable time order [19], a system may serialize them, as under a lock, or proceed optimistically and repair the conflicts after the fact [18, 16]; shared editors and replicated data merge concurrent operations into one state by rules of their own [8, 33]. A simultaneization takes the actions as simultaneous, branches, and reduces by the rules of the game itself, choosing no order. Branchings accumulate until objective information forces their evaluation, a form of lazy evaluation, Section 7.4. The effect is close to what the actors expect: in *SGo* the game behaves like *Go* macroscopically, Section 2.1, though every turn is simultaneous.

The branching paradigm by itself would not surprise computer scientists: it is in use in various contexts, Remark 4.9. What might surprise is that in the *SGo* model the branches interact, see “quantum effects” Section 3.4. Without interactions the branches multiply and evolve side by side, and from the perspective of the agents there are essentially no visible effects: a stone is captured only when it is dead in every branch. The latter is a rough description of the game *DSGo* from Section 6.1. The objective reduction of *SGo* forces effects based on the display; this is where the macroscopic *Go* behavior of Section 2.1 comes from.

2. GAME RULES OF *SGo*

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other’s move on this turn. We allow pass moves as in *Go*. The board the players see and play on we call the **display**. Every stone of the display carries a **time stamp** – the turn of its placement – as part of the display’s data. The capture conditions below read the stamps. Simultaneous action requires modifying the placement and capture rules of *Go*, while removing the Komi, Ko, and suicide rules. We also need to introduce a third type of stone, called **q-stones** (quantum stones), which may be thought of as simultaneously white and black. On the board a q-stone is displayed *red* or, as a visual aid, *blue* when a capture is currently blocked through it, in the sense of the capture conditions below; the colors red and blue are purely visual and carry no rule content.

2.1. Motivation for the rules. Before we proceed to the main rules, we briefly overview the motivation and constraints we impose. We require:

- (1) *SGo* forms a simple simultaneization of *Go*, in the precise sense of Theorem 6.3. Being a simultaneization means that *SGo* naturally maps to the universal simultaneization of *Go*.
- (2) It is played on a standard *Go* board and refereed purely by the display. This in particular requires us to represent simultaneous moves to the same intersection, and we choose to do this by introducing q-stones (visually red or blue).
- (3) No Ko or other non repetition rules.
- (4) We want *SGo* to have comparable game dynamics to the standard *Go*. This is far from the case for the universal simultaneization $\text{Sim}(Go)$ of *Go*, as considered in Section 4.2. For $\text{Sim}(Go)$, games tend to be drawn due to players lacking forcing options.

To achieve these dynamics, we utilize an “objective reduction” process acting on the display each turn (Section 2.5). This is entirely display driven and elementary. In the language of the universal simultaneization, it prunes certain branched histories resulting from simultaneous action, using objective global board information. This has the effect of giving players very forcing options, see Section 3.2 for some examples. But this is not without constraints: we need the game to stay symmetric, while having a well defined deterministic state evolution, as elaborated by Lemma 2.4. We also cannot prune too much, for example in the suicidal capture condition in Definition 2.2 we intentionally preserve certain branching to avoid the Ko type infinite loops, also see Remark 2.3.

Requirement (4) is supported by computer experiments.⁴ The main comparison plays both games on their common rule base – the variant of *Go* fixed before Theorem 6.3 (suicide legal, no superko rule) – under uniformly random play over all legal board moves, passing only when none is available; 1000 games per variant (200 in the last column). On the 5×5 board the game bound of Section 5.1 is set to 40 turns; on 19×19 , *SGo* is left unbounded, while *Go* is stopped at 1500 turns:

	<i>SGo</i> 5×5	<i>Go</i> 5×5	<i>SGo</i> 19×19	<i>Go</i> 19×19
mean margin	13.6	12.0	235.3	68.8
draws	4.6%	5.4%	0.0%	4.5%
games ending on their own	99%	0%	100%	50%
mean length in turns	22.4	39.9	398	1106
q-stones created per game	3.3	–	8.6	–
q-stones on the final display	2.3	–	5.0	–
peak recursion depth	1.1	–	1.0	–

Here the mean margin is the absolute difference of the two area counts, and a game ends on its own when the end of the game rule fires before the bound or cap (the 19×19 figures for *Go* are censored at the cap, so its true mean length is larger still). On 5×5 the margins are nearly identical and the draw rates similar, and the difference is termination: 99% of *SGo* games ended of their own accord within the bound – and with the bound lifted, every one of them does, in 22.4 turns on average – while every classical game was still churning when the bound stopped it.

⁴The simulations and exact computations reported in this paper, together with extensive machine verification of its rules, examples, and arguments – including an exhaustive census of the one-turn faithfulness conditions over every legal classical state of the 3×3 board – were carried out by Claude, an AI system by Anthropic.

On 19×19 the difference is dramatic on both counts: random *SGo* ends by itself on a decisively settled board, with mean margin exceeding half the board, while half of the classical games were still cycling, far from settled, when stopped (every one of the thousand *SGo* games ended on its own).

Disabling the objective reduction – approximating the dynamics of the universal simultaneization of Section 4.2 – collapses the game completely: no stone is then ever removed, the two colors necessarily place equal numbers of stones, and all 500 simulated 5×5 games ended in exact draws. The objective reduction is the entire engine of the dynamics.

One may wonder how informed play changes the above distributions; and also what the depth estimates look like; we partially address this in Sections 7.1 and 7.3.

Finally, the last three rows of the table concern the quantum layer. A finished *SGo* game displays on average about two q-stones on the small board, and five among the 361 points of the large one, so the display stays nearly classical. The recursion depth of a turn is the number of capture reductions (Section 2.5) in its resolution that remove a stone. Its peak over a game averages barely above one, and it never exceeded three in the thousand games at either size. In other words, nearly every turn resolves analogously to *Go*, despite the “quantum engine”.

2.2. Some definitions. Given a board state s , that is any (at the moment no legality restrictions) collection of white, black, and q-stones on the board, each stone implicitly time stamped, let $G(s)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) The display is a board state. From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

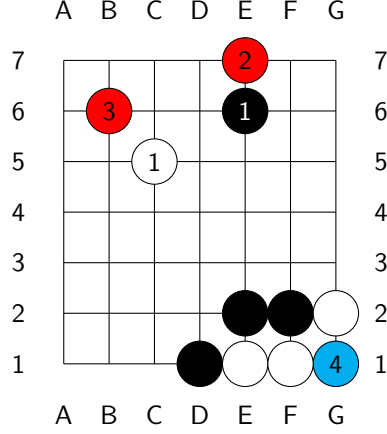
A **unicolor component** C is a connected component of the full (a.k.a. induced) subgraph of $G(s)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.) In this case we call C **black**, respectively **white**. We may less often refer to **q-components**, analogously defined as connected components of q-stones.

As in classical *Go*, a connected component C **has no liberties** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a **trapped component**.

We define stones **adjacent** to C as those stones (not in C) whose corresponding vertices are connected by an edge to C in $G(s)$.

2.3. Stone placement rule of *SGo*. If the players move to the same intersection i on the board, a q-stone is placed at i .

To the right is an example of the placement rule. On turn one the players move to different intersections and both stones are placed as in classical *Go*. On turns two, three and four they move to the same intersection – E7, B6 and G1 respectively – and each conflict yields a q-stone. (One can check that no capture then occurs on turn 4: the blue stone blocks both parts of condition 3 for the white component E1, F1.)



2.4. Capture rule of *SGo*. After both White and Black move on the current turn numbered n , using the stone placement rule above we get a display D .

Definition 2.1. We say that a black, white, or q-component C is **trapped on turn n** if C is trapped and n is the largest stamp among the stones of C and the stones adjacent to C . A stone is called **trapped** on turn n , if it is an element of C as above. We say that a q-stone is **trapped by white respectively black on turn n** (white, respectively black, being the capturing side), if replacing it by a black respectively white stone it is trapped on turn n . For a black respectively white stone, trapped by white respectively black on turn n will simply mean trapped on turn n .

Definition 2.2. A white component C is resolved as **captured on this turn** if the following is satisfied:

- (1) C is trapped on this turn.
- (2) If C is adjacent to a q-stone that is itself adjacent to another trapped component, then that component is white. (Without this condition Lemma 2.4 does not work, and the game evolution will be ill defined.)
- (3) One of the following holds:
 - (a) No black stone or q-stone adjacent to C is trapped by white on this turn.
 - (b) (Suicidal capture) No stone in C is created on this turn. If C is adjacent to a stone trapped on this turn, then the component of that stone contains a stone that is created on this turn and is not a q-stone.

Remark 2.3. The suicidal capture property may appear complicated, but is driven by our requirements, as outlined in Section 2.1. For example, if we remove the part of the condition “is not a q-stone”, then Ko type infinite loops will become possible later. If we remove the condition “no stone in C is created on this turn”, then the proof of Theorem 6.3 will fail.

In the case C is captured:

- The stones of C are removed as in *Go*.
- Any q-stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

A stone converted this way is not considered to be placed or created on this turn; it retains the placement turn of the q-stone it replaces.

If C is a black component the process is naturally mirrored. As far as the rules are concerned this process is simultaneous for all captured components, cf. Lemma 2.4.

2.5. Capture reduction. We call the above the *basic capture reduction* of D , denoted $\widetilde{Cap}_n(D)$. The *capture reduction* is

$$Cap_n(D) = \lim_{k \rightarrow \infty} \widetilde{Cap}_n^k(D),$$

the basic capture reduction iterated until it fixes the display: the sequence is eventually constant, an application removing a stone or fixing the display. In practice the limit is attained at $k = 1$, as is the case in the examples of Section 3 below.

Lemma 2.4. *The basic capture reduction is well defined. Namely, no q-stone is adjacent to components of both colors captured on this turn, so that the displayed conversion of q-stones is unambiguous.*

Proof. Suppose that a q-stone r is adjacent to unicolor components C_0, C_1 , both captured on this turn, with C_0 white and C_1 black. As C_1 is captured, it is in particular trapped. So C_0 is adjacent to the q-stone r , which is itself adjacent to the trapped component C_1 of the opposite color, and Condition 2 fails for C_0 . This contradicts the assumption that C_0 is captured.

Thus, each q-stone adjacent to some component captured on this turn is adjacent to captured components of only one color, and so its conversion is unambiguous. As the removal of stones is likewise unambiguous, each simultaneous resolution step, and hence the basic capture reduction, is well defined. $\square$

If the capture reduction removes no stones, the turn is resolved. Otherwise, define $R^k(D)$ by:

$$\begin{aligned} R^1(D) &= Cap_n(D). \\ R^k(D) &= Cap_{n-k+1}(R^{k-1}(D)). \end{aligned}$$

That is, at stage k the capture conditions of Definition 2.2 are applied with “this turn” read as turn $n - k + 1$. The recursion stops as soon as $R^k(D) = R^{k-1}(D)$, and the turn is then resolved. The total effect is the *objective reduction* (or just OR) of the turn:

$$OR(D) = R^k(D).$$

Remark 2.5. *In practice the resolution of most turns is easy to work out mentally. This is actually quantifiable; in the statistics of Section 2.1 the peak recursion depth never exceeded three and was typically one.*

2.6. End of the game. An SGo game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some black/white stones from the board as dead. (As is common in standard play of Go.) This is the main way of ending the game.
- The display is unchanged after each of two consecutive turns. For example, both players perform a single stone suicide on the same turn twice in a row, or pass twice in a row.
- The display has no empty intersections.

We then decide the winner using Chinese *Go*-style rules. So the winner is decided by comparing the two area counts, which we now explain.

We say that an empty intersection i on the board ***reaches black respectively white*** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is ***surrounded by black*** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a q-stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

- (1) The number of black stones on the board.
- (2) The number of empty intersections surrounded by black.

See Section 3.6 for an example.

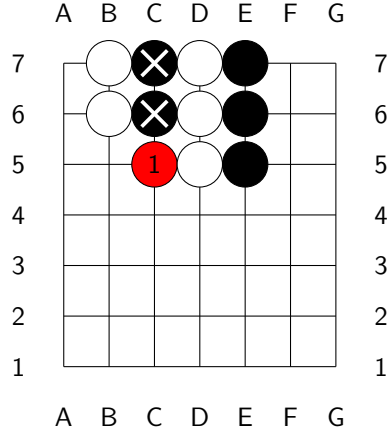
A draw is possible, although compared to chess it is unlikely (for random games on the 19×19 board especially so, as the table in Section 2.1 shows).

3. EXAMPLES

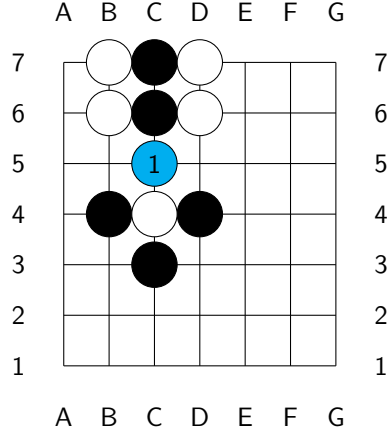
Throughout, a turn's joint move is written m/m' , the move of the first player listed first – in *SGO*, Black's.

3.1. Capture and non capture moves. Here are some examples and non examples of capture.

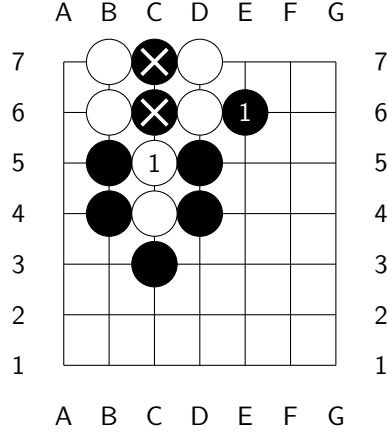
Turn 1 is $C5/C5$. This is a capture move, and C5 will resolve to white at the end of the turn.



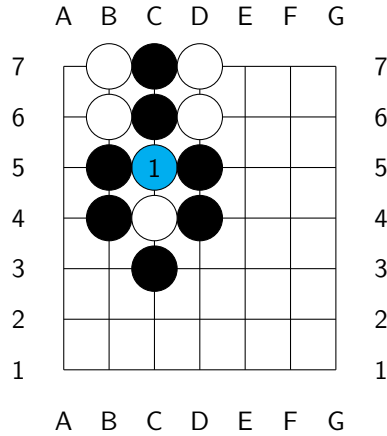
The turn is again $C5/C5$, in another display: not a capture move, as the created q-stone, marked blue, is adjacent to an opposite colored trapped component. This should make intuitive sense since any capture would be ambiguous from a classical *Go* perspective.



Turn 1 is $E6/C5$. White's move is a capture move, as Condition **3b** is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture.

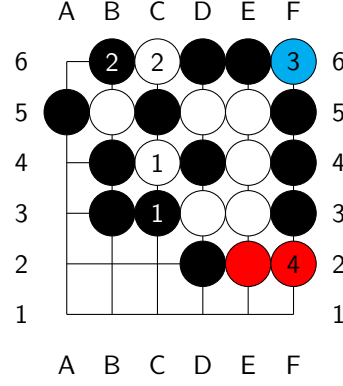


The turn is $C5/C5$: not a capture move, since Condition **3b** is now not satisfied, the stone placed on this turn being a q-stone (marked blue, as the blocker).

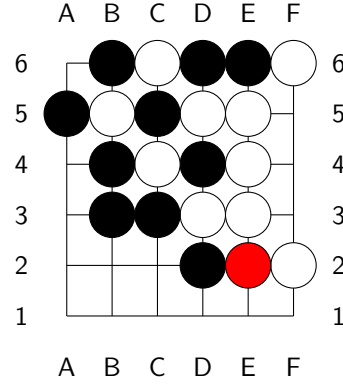


3.2. More complex examples.

In the display on the right, turn 1 is $C3/C4$ – Black attempts capture at C3, White at C4. Nothing is captured on turn 1 according to our rules, and after turn 2 we get a display with captures blocked across turns. The resolution below cascades through the stages. The continuation $B6/C6$; $F6/F6$; $F2/F2$ is indicated. Note that F6 is not a capture for White under our rules, similar to the last example of the previous section; F6 is marked blue, being what blocks the capture of D6 and E6 on this turn. But F2 is a capture move for White since condition [3a](#) is satisfied.

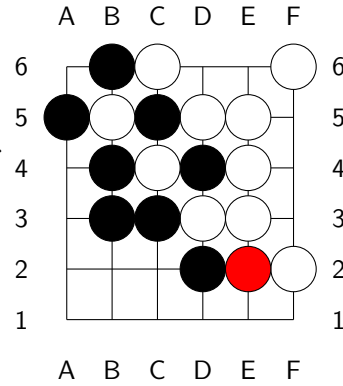


At the end of turn 4, the capture reduction is depicted on the right.



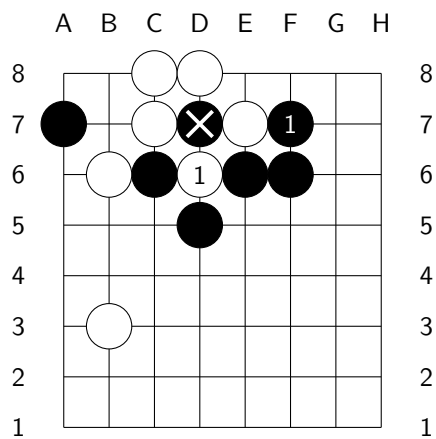
Stage two reduction. Stage three, which applies the capture conditions of turn 2, removes nothing: C5 and B5 each block the other's capture. So the recursion stops, and this is the final resolution of turn 4.

Note that B5, C5, C4 and D4 are all trapped yet uncaptured – an entangled analogue of *seki*, impossible in classical *Go*: each capture is blocked across turns by the others.

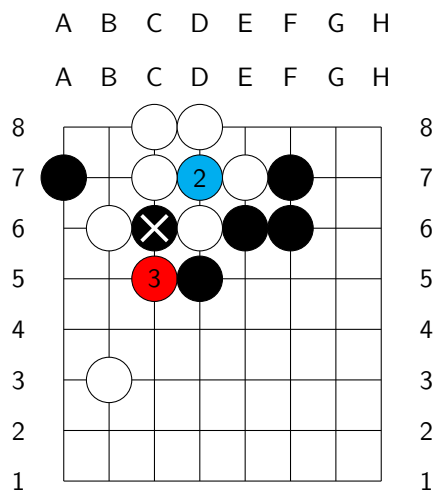


3.2.1. *Ko situation.* Below is a *Ko* type situation. Recall that we do not have a *Ko* rule; none is needed, since the display cannot immediately loop by recapture.

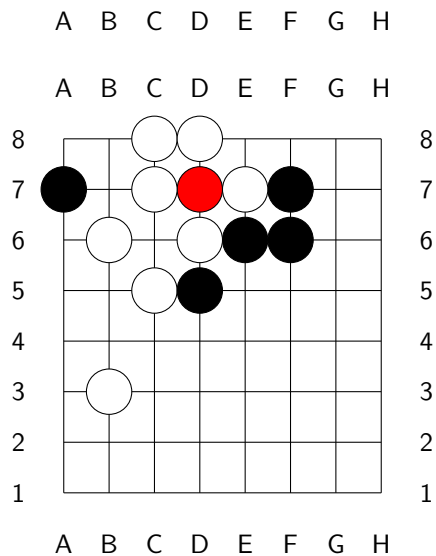
Turn 1 is $F7/D6$; White's $D6$ is a capture move.



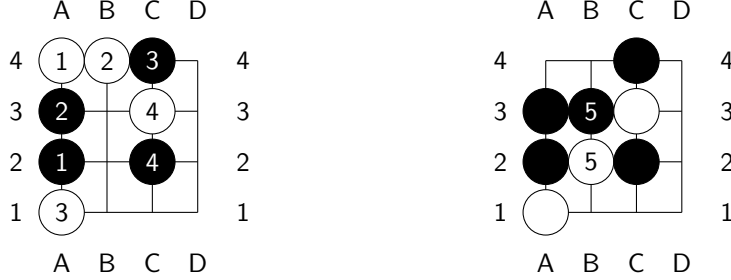
Here is the possible continuation $D7/D7$; $C5/C5$; the blue stone at $D7$ is what blocks Black's recapture of White's $D6$.



At the end of turn 3 we get the following display: the q-stone at $D7$ simply persists. The Ko fight remains open.



3.3. Component clause firing. Here is an example where the component clause of the suicidal capture condition **3b** – that the trapped neighbor’s *component* contain a stone created on this turn – does the work. On the 4×4 board play the conflict-free history $A2/A4$; $A3/B4$; $C4/A1$; $C2/C3$:



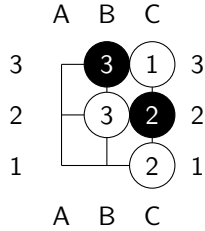
Now play $B3/B2$. Black’s B3 fills the last liberty of the old white pair A4, B4, while White’s B2 fills the last liberty of the black component A2, A3, B3. The black component is trapped by White on this turn, so condition **3a** fails for the pair and the suicidal capture clause governs. The pair touches the trapped component at A3 – an old stone – but the component *contains* this turn’s B3: condition **3b** is met, and the pair is captured, as the right board shows.

This is what the classical semantics demands: the two serializations of the turn agree, and each captures the pair – B3 played first captures it at once, and B2 then lands with the liberty B1; B2 played first, B3 still captures it, the black component surviving on the freed liberties. Formally, condition **3b** is necessitated by condition **4** of Definition 4.13 of simultaneization, see also the preamble of the proof of Lemma 6.4.

3.4. Quantum effects. It helps here to use branch semantics from the formalization of Section 5; the rules themselves never consult that formalization, and the reader may ignore it on first reading.

The branches of the entanglement state can interact through the display; we might call these quantum effects.

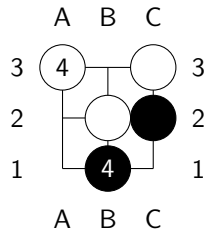
On the 3×3 board play the history $\text{pass}/C3$; $C2/C1$; $B3/B2$, arriving at the display:



Both C3 and C2 are trapped yet uncaptured.

In one branch of the entanglement Black’s B3 has captured C3; in the other White’s B2 has captured C2. This is a standoff on pure black and white stones, with no q-stone anywhere in the history.

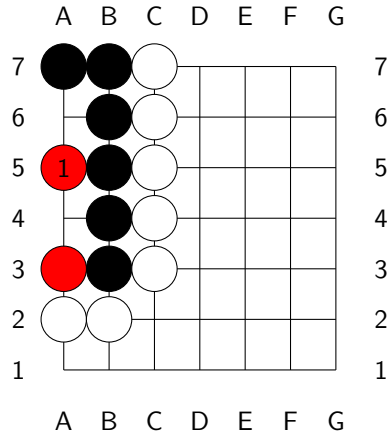
Now play the joint move $B1/A3$. The displayed resolution captures both B3 and C1:



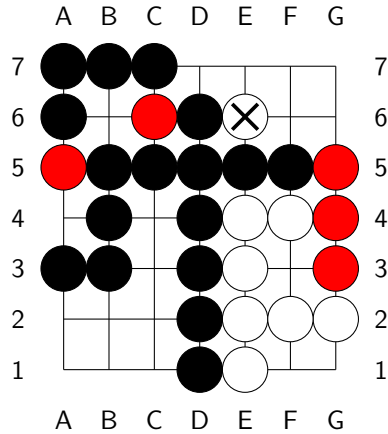
though in each branch only one of them is captured. OR thus performs a joint capture that no serialization of the history performs. In the universal simultaneization of Section 4.2 the branches never interact: its evolution is the union of independent classical evolutions.

3.5. Life and death. Life and death is positionally very similar to *Go*; here is an example.

White cannot kill the black component: A4 and A6 are two eyes – a white stone placed in either is immediately captured, while a simultaneous conflict there yields a q-stone. In *SGo* the component is alive outright: even with Black’s collaboration – Black filling one eye himself, White taking the other – the q-stones at A3 and A5 block every capture. The component survives, wrapped in the entanglement.



3.6. End game scoring. In the diagram below, the players have agreed to end the game. The white stone at E6 is agreed to be dead, and is removed before counting. White has $7+3$ points and Black has $7 \cdot 7 - (7+3) - 5$ points, where the -5 accounts for the five q-stones, which count for neither player.



4. SYMMETRIC SIMULTANEOUS COMBINATORIAL GAMES

We will use the formalism of automata as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games. Such formalisms are standard in several communities: games with simultaneous moves appear as the deterministic case of the stochastic games of Shapley [34], and as the concurrent game structures of computer science [1]. Simultaneous play of combinatorial games as such is studied by Huggan–Nowakowski–Ottaway [13], see also the references therein; there, however, the effect of a pair of simultaneous moves must be specified by each ruleset separately, no general procedure being available. The emphasis here, on simultaneizations of a given sequential game and on their simplicity, appears to be new, and in this setting the pass moves together with the operation $\#$ further below do furnish such a general procedure. The inverse passage, representing a sequential game within an intrinsically simultaneous formalism, is standard in general game playing, and is likewise effected by pass-like moves: there the idle player is required to play an explicit “noop” move, cf. [36].

Definition 4.1. A *combinatorial simultaneous game* (with a pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the *pass move* (it has multi grading both 0 and 1).
- Game states decompose as

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where $B(G)$ is some set.

- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
- (3) If s is not final, $E(s, 0)$ is defined.
- (4) If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' (and if neither m nor m' is 0). We write, recursively, $E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k)$.
- (5) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined, then $E(s, m_0, m_1)$ is defined.

- (6) For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We say that G is **finite** if $S(G)$, $M(G)$ are finite, and all game histories have bounded length, where a game history is a sequence $h = \{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \ E(q_0, m_1, \dots, m_k) \text{ is defined.}$$

It is our convention that $S(G)$ is the set of legally obtainable states, that is states of the form $E(q_0, m_1, \dots, m_k)$, for some m_i .

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 4.2. We say that a combinatorial simultaneous game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{A} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$, and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
- (2) There is a bijection $\mathcal{R} : B(G) \rightarrow B(G)$, satisfying $\mathcal{R}^2 = id$ and $\mathcal{R}(q_0) = q_0$, which we extend to $\mathcal{R} : S(G) \rightarrow S(G)$ by $\mathcal{R}(s, m) = (\mathcal{R}(s), \mathcal{A}(m))$, s.t. for all $s \in S(G)$, all $m \in M(G)$, and for $\mathcal{A}$ as above we have:
 - $\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m)$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
 - $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

4.1. Symmetric combinatorial sequential games. Combinatorial sequential games have a beautiful Conway recursive definition [4]. However, for our purpose, in particular for the universal simultaneization construction, it is convenient to interpret them in terms of finite automata.

Definition 4.3. A **combinatorial sequential game** (with a pass move) G with 2 players P_0, P_1 consists of:

- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move, denoted 0 and called the **pass move**; it has multi grading both 0 and 1.
- A set $C(G)$ of **core states**, with a distinguished initial core state x_0 . The game states are

$$S(G) = C(G) \times \mathbb{Z}_2 \times \mathbb{Z}_3.$$

The set $S(G)$ is $\mathbb{Z}_2$ graded by the middle grading, called the **turn grading**: $S(G) = S_0(G) \sqcup S_1(G)$; the last grading is the **pass grading**. The initial state is $q_0 = (x_0, 0, 0) \in S_0(G)$.

- A partial game evolution function

$$E = E_G : S(G) \times M(G) \rightarrow S(G)$$

with the following properties.

- (1) A state s is **final** (accepting) if $E(s, 0)$ is undefined and in this case $E(s, m)$ is undefined for all m . $F(G)$ denotes the set of final states. States of the form $(x, i, 2)$ are final.

- (2) A move m of grading i is defined only at states of turn grading i and pass grading $j < 2$, and there

$$E((x, i, j), m) = (x', i + 1, 0), \text{ if } m \neq 0$$

$$E((x, i, j), m) = (x, i + 1, j + 1), \text{ if } m = 0$$

with x' , and the definedness of $E((x, i, j), m)$, depending on x and m alone.

- A partition $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G)$ of the final states, understood as states where P_0 wins, P_1 wins or both players draw, respectively.

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

A state is *legally obtainable* if it is of the form $E(q_0, m_1, \dots, m_k)$, for some m_i ; it will be understood that $C(G)$ consists of the core states of legally obtainable states, and $S(G)$ of all the states over them.

Example 4.4. *Note that Go has a pass move, but chess does not; however, chess can be augmented with a pass move, by also adjusting the rules so that king capture is legal and ends the game, while also removing the check rules, and with the game ending in a tie after two consecutive passes. This does not change the early-mid game dynamics at all, but the end game does significantly change, with more tie possibilities (mainly, if a player is in Zugzwang they may just pass). We will call this **augmented chess** denoted by *chess*. In the terms of Definition 4.3, $E(s, 0)$ is defined unless s has a captured king.*

Definition 4.5. We say that a combinatorial sequential game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{A} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$ and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
- (2) There is a bijection $\mathcal{R} : C(G) \rightarrow C(G)$ of the core states, satisfying $\mathcal{R}^2 = id$ and $\mathcal{R}(x_0) = x_0$; extended to the game states by $\mathcal{R}(x, i, j) = (\mathcal{R}(x), i + 1, j)$, it takes $S_0(G)$ to $S_1(G)$, and for all $s \in S(G)$ and all $m \in M(G)$:

$$\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m),$$

whenever both sides are defined, one side being defined if and only if the other is. In particular $\mathcal{R}$ preserves the final states.

- (3) $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Example 4.6. *Of course chess and classical Go are examples of symmetric games. In the case of Go, $\mathcal{A}$ interchanges black and white moves and $\mathcal{R}$ interchanges the colors of the stones of a diagram (it is easy to see that this does result in a legally obtainable core state).*

It will be the assumption from now on that all sequential games that appear are symmetric (with $\mathcal{A}$, $\mathcal{R}$ implicitly fixed).

4.2. Universal simultaneization. Let G be a sequential game (by our standing assumption with a pass move 0 and symmetric).

Definition 4.7. For a state $a = (x, i, j)$ of G with $j < 2$ set

$$\bar{a} = \begin{cases} (x, i+1, 1), & \text{if } j = 0 \\ (x, i+1, 0), & \text{if } j = 1, \end{cases}$$

and for $m \in M(G)$ set

$$a \# m = \begin{cases} E_G(a, m), & \text{if this is defined} \\ E_G(\bar{a}, m), & \text{if this is defined} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

So let m be a non pass move of grading k and $a = (x, i, j)$ a state with $j < 2$. Then $a \# m = E_G((x, k, 0), m)$, each side defined when the other is, by property 2 of Definition 4.3: the move is played at a if $i = k$ and at $\bar{a}$ otherwise. And $a \# 0 = E_G(a, 0)$. At a final state a , $a \# m$ is undefined for every m : E_G is undefined at a final state, and $\bar{a}$, where defined, has the core state of a , so it is final as well, by the same property. A composite $a \# m_0 \# m_1$ therefore passes through no final state.

For a state $b = (x, i, j)$ let $N(b)$, the **position normal form** of b , be $(x, 0, 0)$ for $j < 2$ and $(x, 0, 2)$ for $j = 2$: the gradings are reset, the core state and the finality kept. Two states have the same position normal form exactly when they have the same core state and are both of pass grading 2 or neither is. We write $N(S(G))$ for the set of position normal forms. A position normal form is a state over a legally obtainable core state, though not always a legally obtainable state. Define $P(a, m_0, m_1) \subseteq N(S(G))$, for $a \in S(G)$, by the following prescription in decreasing priority, where each term is understood to be replaced by its position normal form, and the expression is defined if and only if all the terms are defined.

- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1 \# m_0$ (note that if $m_0 = m_1 = 0$ then the two terms coincide, so that $P(a, 0, 0) = a \# 0 \# 0$.)
- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1 \# m_0$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1$.
- $P(a, m_0, m_1)$ is undefined.

We then get a partial function

$$P : S(G) \times M_0(G) \times M_1(G) \rightarrow 2^{N(S(G))},$$

which we call the **serialization map**: it evaluates the turn in its available serializations – both orders of play when both are defined, the degenerate cases otherwise – and collects the outcomes. It is defined exactly when $a \# m_0$ and $a \# m_1$ are: each case has a term through $a \# m_0$ and one through $a \# m_1$, and the fourth case has nothing else.

Definition 4.8. Let G be a sequential game. Its **universal simultaneization** is the simultaneous game $\text{Sim}(G) = G'$ satisfying the following.

- (1) $B(G') = 2^{N(S(G))}$. (The elements s of $\mathfrak{s} \in B(G')$ will be called **state branches**.)
- (2) The initial state $q'_0 = \{q_0\}$.
- (3) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
- (4) Let $\mathfrak{s} \in B(G')$, and $m_i \in M_i(G')$; then $E_{G'}(\mathfrak{s}, m_i)$ is defined iff for each $s \in \mathfrak{s}$, s is not final and $s \# m_i$ is defined.

(5) For $\mathfrak{s}$ and m_i as just above:

$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} P(s, m_0, m_1),$$

(undefined if some $s \in \mathfrak{s}$ is final or some $P(s, m_0, m_1)$ is undefined).

(6)

$$\mathfrak{s} \in F(G') \iff \exists s \in \mathfrak{s} s \in F(G).$$

Note that $\text{Sim}(G)$ is uniquely defined by the above conditions up to the partition of its final states into W_0, W_1, D . The partition plays no role beyond the symmetry of Lemma 4.10; one may take $\mathfrak{s}$ winning for a player exactly when some branch wins for them and no branch wins for the opponent, and drawn otherwise.

By our convention $B(\text{Sim}(G))$ consists of the legally obtainable states. The formulas above make sense at every set of position normal forms, and Definition 4.13 reads $E_{\text{Sim}(G)}$ so, at sets that need not be states of $\text{Sim}(G)$: at the empty set in particular, non final and with every move available, which a branch semantics may reach.

Remark 4.9. *The passage from states to sets of states is a form of the classical subset construction of automata theory [28]. In game theoretic terms, a state of $\text{Sim}(G)$ is a knowledge state: the set of classical states consistent with what the players have observed. Analogous knowledge based constructions are used in the study of games of imperfect information, see for instance [30], [6].*

Lemma 4.10. *$\text{Sim}(G)$ is symmetric.*

Proof. Set $G' = \text{Sim}(G)$, and let $\mathcal{A}, \mathcal{R}$ be the symmetry operators of G . Set $\tilde{\mathcal{A}} : M(G') \rightarrow M(G')$ to just be $\mathcal{A}$. For

$$\{s_0, \dots, s_k\} = \mathfrak{s} \in B(G'),$$

set

$$\tilde{\mathcal{R}}(\mathfrak{s}) = \{N \circ \mathcal{R}(s_0), \dots, N \circ \mathcal{R}(s_k)\}.$$

Clearly, $\tilde{\mathcal{R}}^2 = id$. The computation below is unaffected by the normalization. Indeed, let $a = (x, i, 0)$ and $a' = (x, i + 1, 0)$; for a branch s of pass grading 0, $\mathcal{R}(s)$ and $N(\mathcal{R}(s))$ are such a pair. By the formula for $\#$ after Definition 4.7, $a \# m = a' \# m$ for $m \neq 0$; $a \# 0 = (x, i + 1, 1)$ and $a' \# 0 = (x, i, 1)$ again differ only in the turn grading; and a further pass lands at $(x, i, 2)$ and $(x, i + 1, 2)$, of one position normal form. So $P(a, m_0, m_1) = P(a', m_0, m_1)$. For a final branch s , $\mathcal{R}(s)$ and $N(\mathcal{R}(s))$ are final, and P is undefined at both. For

$$(\mathfrak{s}, m) \in B(G') \times M(G') \subset S(G'),$$

set

$$\tilde{\mathcal{R}}(\mathfrak{s}, m) = (\tilde{\mathcal{R}}(\mathfrak{s}), \tilde{\mathcal{A}}(m)).$$

The needed symmetry properties for the maps $\tilde{\mathcal{A}}, \tilde{\mathcal{R}}$ are then verified using the symmetry properties of $\mathcal{A}, \mathcal{R}$ as follows. For $a \in S(G)$, it is easily seen that $\overline{\mathcal{R}(a)} = \mathcal{R}(\bar{a})$, hence

$$\mathcal{R}(a) \# \mathcal{A}(m) = \begin{cases} E_G(\mathcal{R}(a), \mathcal{A}(m)) = \mathcal{R} \circ E_G(a, m), & \text{if defined} \\ E_G(\mathcal{R}(\bar{a}), \mathcal{A}(m)) = \mathcal{R} \circ E_G(\bar{a}, m), & \text{if defined} \end{cases}$$

and so

$$(4.11) \quad \mathcal{R}(a) \# \mathcal{A}(m) = \mathcal{R}(a \# m),$$

furthermore if one side is defined then so is the other. Suppose that $\mathfrak{s} \in B(G')$ and $m_i \in M_i(G')$; then we get:

$$\begin{aligned} \tilde{\mathcal{R}}(E_{G'}(\tilde{\mathcal{R}}(\mathfrak{s}), \tilde{\mathcal{A}}(m_0), \tilde{\mathcal{A}}(m_1))) &= \tilde{\mathcal{R}} \circ \cup_{s \in \mathfrak{s}} P(\mathcal{R}(s), \mathcal{A}(m_1), \mathcal{A}(m_0)) \\ &= \tilde{\mathcal{R}} \circ \cup_{s \in \mathfrak{s}} \mathcal{R} \circ P(s, m_0, m_1) \text{ using (4.11)} \\ &= \cup_{s \in \mathfrak{s}} P(s, m_0, m_1) \\ &= E_{G'}(\mathfrak{s}, m_0, m_1), \end{aligned}$$

furthermore if one side is defined then so is the other. This verifies property 2 of Definition 4.2 for pair moves at states of $B(G')$; the rest is similar. $\square$

Definition 4.12. Let G_0 be a sequential game and G_1 a symmetric simultaneous game with $M(G_1) = M(G_0)$. The **semi-classical** states of G_1 , with their **associated states** in G_0 , are defined recursively: the initial state of G_1 is semi-classical, with associated state the initial state of G_0 . If s is semi-classical with associated state a , and m_0, m_1 commute at a – both $a \# m_0 \# m_1$ and $a \# m_1 \# m_0$ are defined, with the same position normal form – then $E_{G_1}(s, m_0, m_1)$, if defined, is semi-classical, with associated state the common position normal form of $a \# m_0 \# m_1$ and $a \# m_1 \# m_0$.

Definition 4.13. Let G_0 be a sequential game. We say that G_1 is a **simultaneization** of G_0 if the following is satisfied.

- (1) G_1 is a symmetric simultaneous game.
- (2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
- (3) There is a map $\rho : B(G_1) \rightarrow 2^{N(S(G_0))}$, to the sets of position normal forms – the *branch semantics* – satisfying $\rho(q_0) = q'_0$ and s.t. for $s \in B(G_1)$: whenever the state $E_{G_1}(s, m_0, m_1)$ is defined, so is $E_{\text{Sim}(G_0)}(\rho(s), m_0, m_1)$, read by the formulas of Definition 4.8, and

$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{\text{Sim}(G_0)}(\rho(s), m_0, m_1),$$

the left side possibly empty.

- (4) For every semi-classical s with associated state a (Definition 4.12): $\rho(s) = \{a\}$.

For $a \in S(G_0)$ set

$$\mathbb{I}(a) = \{m \in M(G_0) \mid a \# m \text{ is defined}\},$$

which we call the **interface** of the state a : the set of moves of either player available at a , with the operation $\#$ absorbing the turn order. By Definition 4.7 and property 1 of Definition 4.3, $\mathbb{I}(a)$ is empty exactly when a is final, and contains the pass otherwise. Likewise, for a combinatorial simultaneous game G and $s \in B(G)$ set

$$I(s) = \{m \in M(G) \mid E_G(s, m) \text{ is defined}\},$$

the interface of the state s .

Lemma 4.14. Let G_1 be a simultaneization of G_0 . Then for every non final $s \in B(G_1)$: every branch of $\rho(s)$ is non final, and

$$I(s) \subseteq \bigcap_{a \in \rho(s)} \mathbb{I}(a).$$

That is, a move available at s is available in every branch. Equality of the two sides is precisely the statement that a joint move is available at s if and only if it is available in $\text{Sim}(G_0)$ at $\rho(s)$.

Proof. First, for $m \in M(G_1)$, $E_{G_1}(s, m)$ is defined if and only if the pair move $E_{G_1}(s, m, 0)$ (in the order of the pair prescribed by the grading of m) is defined. Indeed, the latter presupposes the former, and conversely if $E_{G_1}(s, m)$ is defined then so is $E_{G_1}(s, 0)$, as s is not final, and so by the properties of Definition 4.1 the pair move is defined, using that the pass move has both gradings. If some $a \in \rho(s)$ were final, then no joint move would be defined in $\text{Sim}(G_0)$ at $\rho(s)$, by the construction of $\text{Sim}(G_0)$; by condition 3 no joint move would then be defined at s , contradicting the definedness of the pair $(0, 0)$ at the non final s . So every branch is non final. Now if $m \in I(s)$, the pair move is defined at s , hence, by condition 3, defined in $\text{Sim}(G_0)$ at $\rho(s)$, which says exactly that $a \# m$ and $a \# 0$ are defined for every $a \in \rho(s)$, that is $m \in \mathbb{I}(a)$ for every a . The equality statement is the same unwinding read in both directions. $\square$

Definition 4.15. Let G_1 be a simultaneization of G_0 . We say that G_1 is **faithful** at a state $s \in B(G_1)$ if $\rho(s)$ is nonempty and

$$I(s) = \bigcap_{a \in \rho(s)} \mathbb{I}(a).$$

At a non final s this says that exactly the joint moves of $\text{Sim}(G_0)$ at $\rho(s)$ are available at s , Lemma 4.14. At a final s the left side is empty, while a non final branch contributes the pass to the right side; so the equation says that some branch of $\rho(s)$ is final: G_1 may not invent endings where it is faithful. For $d \geq 0$, we say that G_1 is **faithful within distance d of classical play** if it is faithful at every state reachable from a semi-classical state in at most d turns, distance 0 being the semi-classical states themselves. The **faithfulness radius** of G_1 is the supremum of the d such that G_1 is faithful within distance d of classical play, possibly 0 or ∞ : the depth to which its branch semantics stays exact. It is defined when G_1 is faithful within distance 0; otherwise G_1 has no radius.

Lemma 4.16. $\text{Sim}(G)$ is a simultaneization of G , with ρ the identity. Its semi-classical states are singletons. It is faithful at every state, in particular within every distance.

Proof. Condition 1 is Lemma 4.10, condition 2 holds by construction, and condition 3 is an equality for the identity. The initial state is $\{q_0\}$, and at a commuting turn $P(a, m_0, m_1) = \{a'\}$, the common position normal form of the two composites: the semi-classical states are singletons, and condition 4 holds. A state $\mathfrak{s}$ is nonempty, P taking nonempty values. If $\mathfrak{s}$ is final, some branch is, by construction, and both sides of the faithfulness equation are empty. If not, no branch is, and $I(\mathfrak{s}) = \{m \mid s \# m \text{ is defined for every } s \in \mathfrak{s}\} = \bigcap_{s \in \mathfrak{s}} \mathbb{I}(s)$. $\square$

Definition 4.17. Let G_0 be a sequential game. We will say that a simultaneization G_1 of G_0 is a **simple simultaneization of G_0** if for every non final $s \in B(G_1)$ at which a non pass move is available there is a legally obtainable state $a \in S(G_0)$ with

$$\mathbb{I}(a) = I(s),$$

where for s the initial state of G_1 , a can be taken to be q_0 . The **raw radius spectrum** of G_0 is the set of faithfulness radii of its simple simultaneizations, a subset of $\mathbb{N} \sqcup \{\infty\}$; a simultaneization with no radius contributes nothing. The radius spectrum of G_0 , Definition 4.19, is the raw radius spectrum of its minimal quotient.

The radii and the raw radius spectrum are dynamical isomorphism invariants in the sense of the following.

Lemma 4.18. *Let G_0, G'_0 be sequential games, and let φ be a bijection of their core states and of their graded moves; extend it to the game states by $\varphi(x, i, j) = (\varphi(x), i, j)$, and suppose that it intertwines the initial states and the evolutions: $\varphi(E(s, m)) = E(\varphi(s), \varphi(m))$, one side defined if and only if the other. Then:*

- (1) φ intertwines the pass moves, the final states and the position normal forms.
- (2) If H is a simultaneization of G_0 , then renaming each move m of H to $\varphi(m)$ yields a simultaneization H' of G'_0 , isomorphic to H over φ . The faithfulness radii of H and H' coincide, and H' is simple if and only if H is.
- (3) The raw radius spectra of G_0 and G'_0 coincide. The raw radius spectrum is independent of the scoring: repartitioning the final states into W_0, W_1, D does not change it.

Proof. (1) A bijection of graded moves carries $M_0(G_0) \cap M_1(G_0)$ onto $M_0(G'_0) \cap M_1(G'_0)$; each intersection consists of the pass move. Finality is the undefinedness of the pass, which φ preserves. The position normal form resets the gradings, which φ keeps.

(2) By (1), φ intertwines the operation $\#$ and the serialization map P : they are built from the evolutions, the pass moves, the gradings and the position normal form. Hence φ induces an isomorphism $\text{Sim}(\varphi) : \text{Sim}(G_0) \rightarrow \text{Sim}(G'_0)$, acting on a branch set elementwise. Let H' be H with each move m renamed to $\varphi(m)$: the same states, the evolution and the symmetric structure carried along, and the branch reduction $\rho' = \text{Sim}(\varphi) \circ \rho$. The conditions of Definition 4.13 for (H', ρ') are those for (H, ρ) , transported: the semi-classical recursion of H' relative to G'_0 is that of H relative to G_0 , with associated states $\varphi(a)$, so $\rho'(s) = \{\varphi(a)\}$ there. The identity on states, with φ on moves, is then an isomorphism over φ . Interfaces satisfy $I_{H'}(s) = \varphi(I_H(s))$ and $\mathbb{I}(\varphi(a)) = \varphi(\mathbb{I}(a))$; finality in H' is finality in H ; distances of classical play are turn counts. So the faithfulness conditions of Definition 4.15 hold at s in H' if and only if they hold in H , and the radii coincide. For simplicity: a legally obtainable witness a for s in G_0 maps to the legally obtainable witness $\varphi(a)$ in G'_0 , with $\mathbb{I}(\varphi(a)) = I_{H'}(s)$; symmetrically for φ^{-1} . Nothing above reads the partition of the final states into wins and draws.

(3) By (2) applied to φ and to φ^{-1} , the simple simultaneizations of G_0 and of G'_0 realize the same radii; the raw radius spectra coincide. For the scoring statement apply the lemma to the identity bijection. $\square$

The raw radius spectrum is not an invariant of bisimulation, Remark 6.12. The radius spectrum, defined next, is one, Lemma 4.20.

Definition 4.19. Let G, G' be sequential games with $M(G) = M(G')$ as $\mathbb{Z}_2$ graded sets. A **bisimulation** between G and G' is a relation $\sim$ between $C(G)$ and $C(G')$ with $x_0 \sim x'_0$ and such that for $x \sim x'$, all $i \in \mathbb{Z}_2$, $j < 2$ and $m \in M(G)$:

$E((x, i, j), m)$ is defined if and only if $E'((x', i, j), m)$ is, and the core states of the two values are then related. The games are **bisimilar** if a bisimulation between them exists. A bisimulation of G with itself is a bisimulation on G ; the union of all bisimulations on G is one, an equivalence relation, **bisimilarity**, also written $\sim$. The **minimal quotient** $G/\sim$ has the core states $C(G)/\sim$, the initial core state $[x_0]$, and the evolution $E([x], i, j, m) = ([y], i', j')$ for $E((x, i, j), m) = (y, i', j')$; its final states are all drawn. The game G is **minimal** if bisimilarity is the equality. The **radius spectrum** of G is the raw radius spectrum of $G/\sim$, and the **absolute radius** of G is its supremum. So k lies in the radius spectrum of G exactly when $G/\sim$ has a simple simultaneization of radius k ; when G is minimal, exactly when G has one, Lemma 4.20(3).

Lemma 4.20. *Let G be a symmetric sequential game.*

- (1) *$G/\sim$ is a symmetric sequential game, and the projection $C(G) \rightarrow C(G)/\sim$ is a bisimulation.*
- (2) *Bisimilar symmetric sequential games have the same radius spectrum.*
- (3) *If G is minimal, its radius spectrum is its raw radius spectrum.*
- (4) *For a state $a = (x, i, j)$ of G write $[a] = ([x], i, j)$, and let $\pi : 2^{N(S(G))} \rightarrow 2^{N(S(G/\sim))}$ act elementwise, by $a \mapsto [a]$. Then $\pi(\{q_0\}) = \{[q_0]\}$, and $\mathbb{I}([a]) = \mathbb{I}(a)$. A set $\mathfrak{s}$ is final in $\text{Sim}(G)$ exactly when $\pi(\mathfrak{s})$ is in $\text{Sim}(G/\sim)$; a joint move is available at $\mathfrak{s}$ exactly when at $\pi(\mathfrak{s})$, and then*

$$\pi(E_{\text{Sim}(G)}(\mathfrak{s}, m_0, m_1)) = E_{\text{Sim}(G/\sim)}(\pi(\mathfrak{s}), m_0, m_1).$$

So π maps the legally obtainable states of $\text{Sim}(G)$ onto those of $\text{Sim}(G/\sim)$, and the semi-classical states to semi-classical states, the associated state a to $[a]$.

Proof. (1) The evolution of $G/\sim$ is well defined: for $x \sim y$ the moves defined at (x, i, j) and at (y, i, j) agree, and the values have related core states. Its properties 1 and 2 of Definition 4.3 are those of G , read on classes; the scoring is not read by the raw radius spectrum, Lemma 4.18. For the symmetry let $\mathcal{A}$ be that of G and $\mathcal{R}[x] = [\mathcal{R}x]$. This is well defined: the relation $\{(\mathcal{R}x, \mathcal{R}y) \mid x \sim y\}$ is a bisimulation on G , by property 2 of Definition 4.5 – $E((\mathcal{R}x, i, j), m)$ is defined exactly when $E((x, i+1, j), \mathcal{A}m)$ is, with the core state of the value the $\mathcal{R}$ image of that of the other – so it is contained in $\sim$. The properties of Definition 4.5 then pass to the classes. The projection is a bisimulation by construction.

(2) Let $\sim$ be a bisimulation between G and G' , and $\sim_G, \sim_{G'}$ the bisimilarities. Define $\varphi([x]) = [x']$ for any x' with $x \sim x'$. Some x' exists for every legally obtainable core state x : the initial core states are related, and a bisimulation follows every move. The value is independent of the choices: for $x \sim x', y \sim y'$ and $x \sim_G y$ the composite relation $\sim^{-1} \circ \sim_G \circ \sim$ relates x' to y' , and a composite of bisimulations is a bisimulation, so $x' \sim_{G'} y'$. By the symmetric argument φ is injective, and surjective onto the classes of legally obtainable core states. It sends $[x_0]$ to $[x'_0]$ and intertwines the evolutions, by the defining property of $\sim$. So φ , with the identity on moves, is a bijection as in Lemma 4.18, and the raw radius spectra of $G/\sim_G$ and $G'/\sim_{G'}$ coincide.

(3) By minimality the projection is a bijection of core states. With the identity on moves it is as in Lemma 4.18, so the raw radius spectra of G and $G/\sim$ coincide.

(4) By the definition of $G/\sim$, $[E((x, i, j), m)] = E([x], i, j, m)$ for $j < 2$, one side defined exactly when the other. The bar and N act on the gradings alone.

So $[a\#m] = [a]\#m$ and $[N(b)] = N([b])$, one side defined exactly when the other; in particular $\mathbb{I}([a]) = \mathbb{I}(a)$, and a is final exactly when $[a]$ is. The prescription for P decides by definedness, so $\pi(P(a, m_0, m_1)) = P([a], m_0, m_1)$, one side defined exactly when the other. Availability, the evolution and finality in Sim read these alone, Definition 4.8: the claims on $\mathfrak{s}$ follow. The image of the legally obtainable states of $\text{Sim}(G)$ consists of legally obtainable states of $\text{Sim}(G/\sim)$, contains the initial state and is closed under the evolution: it is the set of legally obtainable states of $\text{Sim}(G/\sim)$. A turn commuting at a commutes at $[a]$; so along the recursion of Definition 4.12, π carries semi-classical states to semi-classical states, a to $[a]$. $\square$

5. FORMALIZING SGo

A state s of SGo is a pair $(D_s, \rho(s))$. The first member is the display, as maintained by the rules of Section 2, its time stamps included. The second is the **entanglement state** $\rho(s)$, briefly the *entanglement*: a finite, possibly empty, set of classical Go states, called **branches** – the classical states that the play so far may have produced. Note that this is decorative: it is extra data we choose to baggage for reasons that will be apparent soon. The game rules themselves are completely display driven; the player does not care about this baggage. Later we introduce a variant game $DSGo$ whose game evolution does consult the entanglement.

The initial entanglement state is $\{q_0\}$, the singleton of the initial state of Go : the empty diagram, Black to move. On a turn m_0/m_1 , the entanglement set evolves by $\text{Sim}(Go)$ evolution, followed by pruning of the branches incompatible with the display. **Compatible** means: every stone of a branch stands on an occupied intersection of the display, and matches the color of the displayed stone there, a q-stone matching either color. Let $\mathcal{P}_D$ denote this pruning operator on entanglement sets with respect to display D . Then:

$$E_{SGo}(D_s, \rho_s), m_0, m_1) = (D_{s'}, \mathcal{P}_{D_{s'}}(E_{\text{Sim}(Go)}(\rho(s), m_0, m_1))),$$

where $D_{s'}$ is the display after the turn resolves, by the rules of Section 2.

Lemma 5.1. *SGo is an ssc game, that is a symmetric simultaneous combinatorial game.*

Proof. Set $G = SGo$. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description, with the pass move 0 the standard pass move.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where an element of $B(G)$ is the full data maintained by the rules of Section 2: the display with its time stamps, together with the entanglement state of Section 5 – a finite set of states of Go in position normal form, Section 4.2. The initial state is the empty display with the entanglement $\{q_0\}$, for q_0 the initial state of Go .

The evolution map E is then naturally defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are moves to empty intersections of the display, then $E((s, m_0), m_1) \in B(G)$ is defined to be the state produced by the rules of Section 2: the display resolves by the placement rule and the objective reduction. Each branch evolves under the serialized moves, and the branches incompatible with the resolved display are discarded.

To see that G is symmetric, let $\mathcal{R} : B(G) \rightarrow B(G)$ be the map that interchanges the black and white colors of the stones in every branch (on the display this interchanges black and white and leaves q-stones invariant). And let $\mathcal{A} : M(G) \rightarrow M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \rightarrow M_1(G)$, which then induces $\mathcal{A}$. Then clearly the conditions of Definition 4.2 will be satisfied. $\square$

Remark 5.2. *We have not formalized $F(SGo)$: ending a game by agreement is not an automaton notion. This caveat is however minor, and is already present in classical Go. For the theorems of Section 6 the automaton convention is fixed there: a state is final after a double pass turn, or if only pass is available at it.*

5.1. Solvability. A theorem of Zermelo [32] says that for every finite sequential game either Black or White has a winning strategy or there is a draw strategy. Clearly SGo cannot have a winning strategy: mirroring a winning strategy via the symmetry operators would yield one for the opponent, and both players cannot win. Furthermore, unless we put an upper bound on the length of a game, or on the number of state repetitions, an SGo game can be infinitely long.⁵ So we will assume there is such a bound, which will make SGo a finite game.

One may ask whether SGo has a *strategy to draw*: a pure strategy guaranteeing at least a draw. On the 2×2 board it does, with a caveat: here we drop the game length bound and count an unending game as drawn. In this sense, exact computation shows that both players can force at least a draw. Neither player can force the game to end: the drawing strategies are locks, not finishes. But already on the 3×3 board, with any bound between 6 and 8, neither player has such a strategy: a player following any fixed pure strategy can be made to lose. Optimal play is thus necessarily mixed from the 3×3 board on, and we expect the same on all larger boards – the mixed equilibrium of Theorem 5.3 below is not a formality.

Theorem 5.3. *For any assignment of payoffs to a win, a loss and a draw, a finite combinatorial simultaneous game G (Definition 4.1) has a Nash equilibrium.*

Proof. For a non final $s \in B(G)$ set

$$I_i(s) = \{m \in M_i(G) \mid E(s, m) \text{ is defined}\},$$

which is nonempty as it contains the pass move 0.

A **pure strategy** of player P_i assigns to each non final $s \in B(G)$ a move in $I_i(s)$, so that the set of pure strategies is the finite set

$$\Sigma_i = \prod_s I_i(s),$$

the product over the non final elements of $B(G)$. A pair of pure strategies determines a play of the game, and so a payoff, by finiteness of G .⁶ This presents G in normal form: a finite two-player game with the strategy sets Σ_0, Σ_1 . The mixed strategies of player P_i are then $\mathcal{P}_i = \Delta(\Sigma_i)$, the probability distributions over Σ_i . The theorem is now a direct application of the Nash existence theorem [24]. $\square$

Corollary 5.4. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

⁵This is also true for Go and chess.

⁶Possibly the finiteness condition can be relaxed.

Proof. Assign 1 for a win, -1 for a loss and 0 for a draw. By the normal form presentation in the proof of Theorem 5.3, G is a zero sum matrix payoff game, with the rows and columns of the payoff matrix indexed by the pure strategies Σ_0, Σ_1 . By the classical theory of matrix payoff games (see for instance [25]), all Nash equilibria have the same expected payoff for P_0 , namely the value v of the matrix game, and an equilibrium strategy of P_0 guarantees P_0 an expected payoff of at least v against every strategy of P_1 . So it is enough to show that $v = 0$; that a symmetric zero sum game has value zero is classical, cf. [11], and for completeness we give the argument.

Suppose otherwise, and suppose without loss of generality that $v > 0$, so that for an equilibrium joint strategy

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

P_0 has expected payoff $v > 0$.

Let $\mathcal{R}, \mathcal{A}$ be the symmetry operators of G , Definition 4.2. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where P_1 has expected payoff v , that is with P_0 expected payoff $-v < 0$. As the latter must again be v , we have a contradiction. $\square$

6. MAIN THEOREMS

6.1. The decoherent variant $DSGo$. We now introduce a theoretical variant $DSGo$: SGo with the objective reduction preempted by decoherence. Each turn, the stones are placed by the placement rule of SGo , and nothing else is played out on the display. The resulting state is then postprocessed as follows. The display is measured against the branch evolution of the turn; the entanglement does the measuring and the removing. A state of $DSGo$ carries the same data as one of SGo – a display together with its entanglement state $\rho(s)$ – and moves are available as in SGo .

The serialization map extends to sets of classical states elementwise,

$$P(\mathfrak{B}, m_0, m_1) = \bigcup_{a \in \mathfrak{B}} P(a, m_0, m_1);$$

we call $P(\rho(s), m_0, m_1)$ the *decoherence* of the turn m_0/m_1 at s .

For an intersection i and a classical state b , let $ev_i(b) \subseteq \{\text{black}, \text{white}\}$ be the color of the stone of b at i , the empty set if i is empty in b . Extend ev_i to a set of classical states by union, and to a display, or a state carrying one, by reading the display, a q-stone contributing $\{\text{black}, \text{white}\}$. The *diagram* of x – a classical state, a display, or a state carrying one – is the function $\text{diagram}(x) : i \mapsto ev_i(x)$ on the intersections of the board. Equality of diagrams is equality of stone patterns, q-stones included.

Next, the *postprocessing* $\Delta(s, \mathfrak{M})$ of a state s of $DSGo$ by a finite set $\mathfrak{M}$ of classical states, anchored at the current turn, is defined as follows. Set D_0 the display of s and $\mathfrak{M}_0 = \mathfrak{M}$. Proceed recursively; for $j \geq 0$, postprocessing round j removes from D_j the stones at the intersections i with $ev_i(\mathfrak{M}_j) = \emptyset$. A q-stone of D_j created on this turn, at an intersection i with $ev_i(\mathfrak{M}_j) = \{c\}$, becomes a c stone. This gives D_{j+1} .

Based on D_{j+1} the entanglement is then re-cut:

$$\mathfrak{M}_{j+1} = \{b \in \mathfrak{M}_j \mid \forall i : ev_i(D_{j+1}) = \emptyset \implies ev_i(b) = \emptyset\}.$$

The recursion stops at the first j with $D_{j+1} = D_j$; such a j exists as infinite descent is impossible.

Then $\Delta(s, \mathfrak{M})$ has display D_{j+1} and entanglement state $\mathfrak{M}_{j+1}$. The evolution of $DSGo$ is the postprocessed placement:

$$E_{DSGo}(s, m_0, m_1) = \Delta(\text{Pl}(s, m_0, m_1), P(\rho(s), m_0, m_1)),$$

where $\text{Pl}(s, m_0, m_1)$ is s with the turn's stones placed by the placement rule of SGo and no objective reduction applied. The preemption is forced. Were the objective reduction to act first, its captures standing, every element of the decoherence could be lost in the re-cut; faithfulness would then fail at distance two.

The postprocessing loses no element of the decoherence: an element of $\rho(s)$ occupies only display-occupied intersections – the re-cut keeps nothing else – a placement empties no intersection, and classical play otherwise only removes stones; so every element of the decoherence satisfies the hypothesis of Lemma 6.1(1) below, and survives. Hence $\rho \circ E_{DSGo} = E_{\text{Sim}(Go)} \circ \rho$: the entanglement component of a $DSGo$ history is a $\text{Sim}(Go)$ history. $DSGo$ referees $\text{Sim}(Go)$ on a display, differing from it only in scoring: the winner is decided as in SGo , Section 2.6, from the display.

The construction interpolates. For $k \geq 1$, the delayed game k - $DSGo$ stays the execution of the objective reduction for k turns. Formally, the objective reduction $D \mapsto OR(D)$ of Section 2.5, anchored at the current turn, only removes stones and converts q-stones. The *verdict* $\text{ver}(D)$ is the difference: the set of stones of D removed in $OR(D)$, and the set of q-stones of D made definite in $OR(D)$, with their acquired colors. For a verdict v , the execution $v \cdot D'$ removes and recolors accordingly the stones of D' that v names, a stone being named by its intersection and its time stamp; named stones no longer standing in D' are ignored.

A state of k - $DSGo$ is a tuple $(D, \mathfrak{M}, v_1, \dots, v_k)$: a display, an entanglement state, and the verdicts of the last k turns, v_1 the oldest. The initial state is that of $DSGo$ with all k verdicts empty. Moves, scoring, and finality are those of $DSGo$, read from D and $\mathfrak{M}$. The evolution is

$$\begin{aligned} E_k((D, \mathfrak{M}, v_1, \dots, v_k), m_0, m_1) &= (D', \mathfrak{M}', v_2, \dots, v_k, \text{ver}(D')), \\ (D', \mathfrak{M}') &= \Delta(\text{Pl}((v_1 \cdot D, \mathfrak{M}), m_0, m_1), P(\mathfrak{M}, m_0, m_1)) : \end{aligned}$$

the oldest verdict is executed, the turn is then a turn of $DSGo$, and the new verdict is stored. Verdicts still pending when the game ends are discarded: the display is scored as it stands. As $k \rightarrow \infty$ no verdict ever executes, and the game is $DSGo$; SGo , which plays its OR at once and carries no decoherence, stands at the origin of the hierarchy.

Lemma 6.1. *Let s be a state of $DSGo$ and $\mathfrak{M}$ a finite set of classical states.*

- (1) *If $b_0 \in \mathfrak{M}$ satisfies, for every intersection i , $ev_i(s) = \emptyset \implies ev_i(b_0) = \emptyset$, then $b_0 \in \rho(\Delta(s, \mathfrak{M}))$.*
- (2) *For every intersection i : $ev_i(\Delta(s, \mathfrak{M})) = \emptyset \iff ev_i(\rho(\Delta(s, \mathfrak{M}))) = \emptyset$.*

Proof. In the rounds $(D_j, \mathfrak{M}_j)$ defining $\Delta(s, \mathfrak{M})$ – the recoloring does not change occupancy and plays no role – for (1) we show by induction on j : $b_0 \in \mathfrak{M}_j$, and $ev_i(D_j) = \emptyset \implies ev_i(b_0) = \emptyset$ for all i . For $j = 0$ this is the hypothesis. For the step: an intersection emptied by round $j + 1$ has $ev_i(\mathfrak{M}_j) = \emptyset$, and $ev_i(b_0) \subseteq ev_i(\mathfrak{M}_j)$ since $b_0 \in \mathfrak{M}_j$. With the hypothesis, $ev_i(D_{j+1}) = \emptyset \implies ev_i(b_0) = \emptyset$, hence $b_0 \in \mathfrak{M}_{j+1}$.

For (2): let D and $\mathfrak{M}_*$ be the final display and entanglement, so $\mathfrak{M}_* = \rho(\Delta(s, \mathfrak{M}))$. If $ev_i(D) = \emptyset$ then $ev_i(b) = \emptyset$ for every $b \in \mathfrak{M}_*$, by the re-cut; so $ev_i(\mathfrak{M}_*) = \emptyset$. Conversely, a stone of D at an intersection i with $ev_i(\mathfrak{M}_*) = \emptyset$ would be removed by a further round, and the repetition has stabilized. $\square$

Let Go be the classical game with Chinese style area scoring and standard pass move, with the following stipulations: suicide is a legal move, as for example in the Tromp–Taylor rules [38, 37], there is no Ko or superko rule, and the game ends by a double pass: a state of Go is final exactly after two consecutive passes, Definition 4.3. The same convention is fixed for SGo and its variants below: a state is final exactly after a double pass turn. Of the end of the game rules of Section 2.6 this keeps the double pass alone, ending by agreement not being an automaton notion, Remark 5.2. The board is any square board of side at least two.⁷

In the terms of Definition 4.3: the core states of Go are the classical diagrams, the initial core state the empty diagram, $E(s, 0)$ is defined at every state (x, i, j) with $j < 2$, and the non pass moves act as resolved below. A non pass move plays to an empty intersection; a pass changes nothing. For a classical diagram B and a non pass move m whose placement intersection is empty in B , write $B + m$ for B with the stone of m entered at its placement intersection – the *placed diagram*. Components and liberties are read off a diagram. For a legal classical state b with $b\#m$ defined, write $B = \text{diagram}(b)$; classical play resolves:

- (1) The components captured by the move are the opposite colored components of $B + m$ with no liberties in it.
- (2) The component of the placed stone suicides if and only if it has no liberties in $B + m$ with the captured components emptied.
- (3) The diagram of $b\#m$ is $B + m$ with the captured and suicided components emptied; in particular the move fills no intersection but its placement intersection.

We write $B\#m$ for this diagram: $\text{diagram}(b\#m) = \text{diagram}(b)\#m$.

Lemma 6.2. *$DSGo$ and $k\text{-}DSGo$, $k \geq 1$, are symmetric simultaneous combinatorial games.*

Proof. The conditions of Definition 4.1 hold as for SGo , Lemma 5.1: the moves are those of SGo , and the final states are the states after a double pass turn, by the convention above.

For the symmetry, let $\mathcal{A}$ interchange black and white moves. Let $\mathcal{R}$ interchange black and white on the display, q-stones and stamps kept; act on the entanglement as $\tilde{\mathcal{R}}$ in the proof of Lemma 4.10, for $G = Go$; and interchange the acquired colors in each stored verdict. Then $\mathcal{A}^2 = id$, $\mathcal{R}^2 = id$, and $\mathcal{R}(q_0) = q_0$: the empty display, the branch q_0 of Go and the empty verdicts are fixed. By the order invariance of E , the last property of Definition 4.1, property 2 of Definition 4.2 reads: at $s \in B(G)$, $\mathcal{A}(m)$ is available at $\mathcal{R}(s)$ exactly when m is available at s , and

$$\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m_1), \mathcal{A}(m_0)) = E(s, m_0, m_1),$$

⁷The 1×1 board is a genuine exception: there the joint move to the sole intersection commutes – each lone stone is a suicide – while the display keeps a q-stone, and faithfulness fails at distance zero. Theorem 6.3 is stated for side at least five: the witness for the radius of $k\text{-}DSGo$ pads with turns in a corner away from its stones. The clauses on SGo and $DSGo$ hold from side two on, the SGo witness using side at least three, and the 2×2 board computer exhaustion, which finds failures at distance one for SGo and at distance two for $1\text{-}DSGo$.

one side defined exactly when the other.

Availability: a non pass move is available exactly when its placement intersection is empty on the display, which $\mathcal{R}$ keeps; the pass is available at every non final state, and $\mathcal{R}$ preserves finality, below.

The evolution of $DSGo$ is $\Delta(\text{Pl}(s, m_0, m_1), P(\rho(s), m_0, m_1))$; that of $k\text{-}DSGo$ executes the oldest verdict first and stores $\text{ver}(D')$ last. Each factor commutes with the interchange.

- (1) $\text{Pl}(\mathcal{R}(s), \mathcal{A}(m_1), \mathcal{A}(m_0)) = \mathcal{R}(\text{Pl}(s, m_0, m_1))$: each side is $\mathcal{R}(s)$ with a black stone at the intersection of m_1 and a white stone at that of m_0 , a q-stone if these coincide, stamped with the turn.
- (2) $P(\tilde{\mathcal{R}}(\mathfrak{M}), \mathcal{A}(m_1), \mathcal{A}(m_0)) = \tilde{\mathcal{R}}(P(\mathfrak{M}, m_0, m_1))$, one side defined exactly when the other: this is the computation closing the proof of Lemma 4.10, for $G = Go$, a symmetric sequential game by Example 4.6.
- (3) $ev_i(\tilde{\mathcal{R}}(\mathfrak{M}))$ is $ev_i(\mathfrak{M})$ with the colors interchanged. So a postprocessing round of $(\mathcal{R}(s), \tilde{\mathcal{R}}(\mathfrak{M}))$ removes the stones a round of $(s, \mathfrak{M})$ removes, converts the q-stones it converts, to the interchanged colors, and re-cuts the same branches: $\Delta(\mathcal{R}(s), \tilde{\mathcal{R}}(\mathfrak{M})) = \mathcal{R}(\Delta(s, \mathfrak{M}))$.
- (4) Definition 2.2 states the capture of a black component as that of a white one with the colors interchanged, the conversion of the adjacent q-stones included. So the basic capture reduction commutes with $\mathcal{R}$, hence so does OR , and $\text{ver}(\mathcal{R}(D)) = \mathcal{R}(\text{ver}(D))$.
- (5) A verdict names stones by intersection and stamp: $\mathcal{R}(v) \cdot \mathcal{R}(D) = \mathcal{R}(v \cdot D)$.

By (1)–(3), $E_{DSGo}(\mathcal{R}(s), \mathcal{A}(m_1), \mathcal{A}(m_0)) = \mathcal{R}(E_{DSGo}(s, m_0, m_1))$, one side defined exactly when the other; by (1)–(5), likewise for E_k . Apply $\mathcal{R}$.

Scoring and finality: the area counts of $\mathcal{R}(D)$ are those of D with the players interchanged, so $\mathcal{R}$ maps $W_0(G)$ onto $W_1(G)$ and the draws onto the draws. A double pass turn is sent to a double pass turn, $\mathcal{A}(0) = 0$; so $\mathcal{R}$ preserves the final states. $\square$

Theorem 6.3. *On a board of side at least five: SGo , the delayed games $k\text{-}DSGo$ for $k \geq 1$, and $DSGo$ are simple simultaneizations of Go . Their faithfulness radii are, respectively, zero, exactly k , and infinity. In particular the raw radius spectrum of Go is all of $\mathbb{N} \sqcup \{\infty\}$, and so is its radius spectrum, Go being minimal (Lemma 6.13).*

Proof. Call the games in the hypothesis the SGo family. Condition 1 is Lemma 5.1 for SGo and Lemma 6.2 for the rest of the family; condition 2 holds by construction.

Let $\rho : B(SGo) \rightarrow 2^{N(S(Go))}$ send a state to its entanglement component: the entanglement state is a finite set of states of Go in position normal form – the terms of P are normalized. Then $\rho(q_0) = \{q_0\} = q'_0$. For $DSGo$ and the delayed games, ρ likewise reads the entanglement component $\mathfrak{M}$. The branch evolution of Section 5 is precisely the evolution by the serialization map P : playing the two moves in both orders, with a move to an occupied intersection forfeited, is the prescription defining P . So $E_{SGo}(s, m_0, m_1)$ is obtained by applying the OR prunings to $E_{\text{Sim}(Go)}(\rho(s), m_0, m_1)$, and the inclusion in condition 3 holds by construction, similarly for the other members of the family.

For the definedness direction of condition 3: a move of SGo is available if and only if its placement intersection is empty on the display; a branch kept by the reduction has all its stones within the display, a branch occupying a display-empty

intersection being incompatible with the display and discarded; so a display-empty intersection is empty in every branch. Every branch of a non final s is non final: a final branch, of pass grading 2, arises only from $P(b, 0, 0)$, and the double pass turn ends the game, by the convention above. So the joint move is available in $\text{Sim}(Go)$ at $\rho(s)$. This is similar, for the rest of the family.

We verify condition 4 following the recursion of Definition 4.12. At the initial state it is the normalization $\rho(q_0) = q'_0$. For the step: suppose s is semi-classical with associated state a and $\rho(s) = \{a\}$, and let s' be the state after a turn m_0/m_1 commuting at a , with a' its associated state (Definition 4.12). It is enough to verify that $\rho(s') = \{a'\}$, ρ the branch semantics of the respective game.

For $\text{Sim}(Go)$ this is immediate: ρ is the identity, and at a commuting turn the two composites have the common position normal form a' :

$$E_{\text{Sim}(Go)}(\{a\}, m_0, m_1) = P(a, m_0, m_1) = \{a'\}.$$

For $DSGo$, we have observed in Section 6.1 that $\rho \circ E_{DSGo} = E_{\text{Sim}(Go)} \circ \rho$, hence it follows by the above.

For $DSGo$ we also have

$$\text{diagram}(s) = \text{diagram}(a) \implies \text{diagram}(s') = \text{diagram}(a').$$

The equality we need to show is: $ev_i(s') = ev_i(a')$ at every intersection i . On the empty intersections this is Lemma 6.1(2), applied to $\rho(s') = \{a'\}$. Concretely, the first postprocessing round empties the intersections of the placed display that are empty in a' – the stones the turn removes in classical play – and the second round changes nothing. We show that on the occupied intersections the colors are those of a' . A stone of the turn is placed as in a' : the moves of a commuting turn are to distinct intersections, Step 0 of the proof of Lemma 6.4 below, so no q-stone is created. An older stone has its color by the hypothesis, a' keeping the stones of a it does not remove.

For $k-DSGo$ the recursion carries two more invariants: $\text{diagram}(s) = \text{diagram}(a)$, and the stored verdicts of s are empty. At the initial state all three hold. For the step: the executed verdict is empty, so the display is unchanged, and the turn is a turn of $DSGo$; $\rho(s') = \{a'\}$ and $\text{diagram}(s') = \text{diagram}(a')$ as above. The new verdict is empty. The equality $\text{diagram}(s') = \text{diagram}(a')$ reads: $ev_i(s') = ev_i(a')$ at every intersection i . No $ev_i(a')$ is the pair, so no q-stone stands on the display of s' . Components and liberties are read off the diagram, and a' is legal: every component of the display has a liberty, no component is trapped, and OR removes and recolors nothing.

For SGo the recursion carries the invariant $\text{diagram}(s) = \text{diagram}(a)$. It propagates by Lemma 6.4 below: $\text{diagram}(s') = \text{diagram}(a')$. The branches of the turn are $P(\{a\}, m_0, m_1) = \{a'\}$. The equality $\text{diagram}(s') = \text{diagram}(a')$ reads: $ev_i(s') = ev_i(a')$ at every intersection i . So every stone of a' occupies an intersection occupied on the display of s' , with its color: a' is compatible, and survives the pruning: $\rho(s') = \{a'\}$.

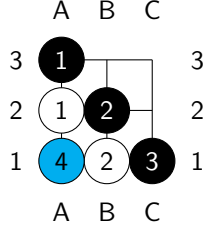
To verify the simplicity property: since in our version of Go suicide is a legal move and there is no Ko rule, for any non final state a of Go the interface $\mathbb{I}(a)$ consists of the pass move and the moves of either player to the empty intersections of a , that is of its diagram. The same holds for SGo : for a non final $s \in B(SGo)$, $I(s)$ consists of the pass move and the moves of either player to the empty intersections of $\text{diagram}(s)$. Let a non pass move be available at s ; its placement intersection i_0

is then empty in $\text{diagram}(s)$. So it suffices to exhibit a legally obtainable state $\phi(s)$ of Go with the same empty intersections as $\text{diagram}(s)$. Let $\phi(s)$ have the diagram obtained by replacing every stone of $\text{diagram}(s)$ by a black stone. Since i_0 is empty, every component of this diagram, and of every all black diagram within it, has a liberty. So $\phi(s)$ is legally obtainable: Black places the stones one at a time, White passing, no placement being a capture or a suicide.

The simplicity argument applies verbatim to the rest of the family. For these games, the inclusion of condition 3 is the re-cut of Δ , by definition; and a surviving element occupies only display-occupied intersections, which gives the definedness direction as before.

It remains to compute the radii of the family.

For SGo : at a semi-classical state s with associated state a , $\text{diagram}(s) = \text{diagram}(a)$ and $\rho(s) = \{a\}$, as established above; so $I(s) = \mathbb{I}(a)$ when s is not final. By the convention fixed above s is final exactly after a double pass turn, and so is its associated state; both interfaces are then empty. Thus SGo is faithful within distance zero. It is not faithful within distance one: reach, by commuting turns, black stones on A3, B2, C1 and white stones on A2, B1, and play the conflict A1/A1, reaching a state s' :



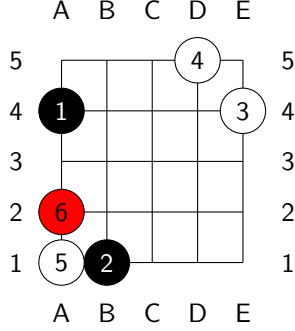
In one order Black's A1 captures A2 and B1; in the other White's A1 is a suicide, and Black's A1 then lands on the emptied intersection. Both orders produce the same state, with A2 and B1 empty; the entanglement of s' is its singleton. On the display, the created q-stone is trapped by White on this turn, so Condition 3a fails for each white stone; and Condition 3b fails on its component clause, the q-stone's component carrying no stone of the turn that is not a q-stone. So the OR removes nothing (compare the last example of Section 3.1). So A2 and B1 stand on the display yet are empty in every element of the entanglement: $I(s') \subsetneq \bigcap_{b \in \rho(s')} \mathbb{I}(b)$. The radius of SGo is zero.

For the radius of $DSGo$: by the convention fixed above a state s' is final exactly after a double pass turn, and then every element of $\rho(s')$ is final, $P(b, 0, 0) = \{b \# 0 \# 0\}$ for every branch b : both sides of the faithfulness equation are empty.

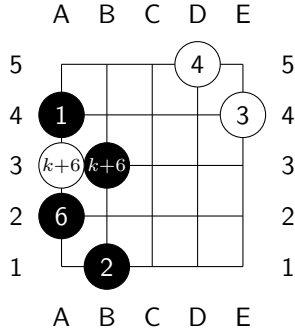
For a non final s' , reached from a reachable state s by a turn: $\rho(s')$ is the entire decoherence $P(\rho(s), m_0, m_1)$, as observed in Section 6.1, and it is nonempty. By Lemma 6.1(2), an intersection is empty on the display of s' if and only if it is empty in every element of $\rho(s')$, that is $I(s') = \bigcap_{b \in \rho(s')} \mathbb{I}(b)$: faithfulness at s' . So $DSGo$ is faithful at every state reachable in play, and its faithfulness radius is infinite.

For the radius of k - $DSGo$: within distance k of classical play, every verdict that executes is one of the k verdicts stored at the semi-classical state, the j -th turn after it executing the j -th oldest; all are empty, as shown above. So every turn within this distance is a turn of $DSGo$, and the preceding argument applies: k - $DSGo$ is faithful within distance k .

It is not faithful within distance $k + 1$. Reach, by commuting turns, black stones on A4 and B1 and white stones on A1 and on the two neighbors of the corner opposite A1 – E4 and D5 on the 5×5 board – and play the conflict A2/A2:



The conflict resolves by the lowest priority case of the serialization map P : the entanglement becomes $\{a_0, a_1\}$, where in a_0 Black's A2 has captured A1, and in a_1 White's A2 has joined A1. The verdict of the new display – A1 removed, the q-stone made definite as black – is stored. Then, for $k - 1$ turns, Black plays the corner opposite A1 and White passes. In every branch the move is a single stone suicide, the neighbors of the corner being white, and the branch is unchanged. On the display the placed stone stands in no branch: the first round of Δ removes it, and display and entanglement return to those after the conflict. Each of these turns executes an empty verdict, stored before the conflict. Now play B3/A3, reaching a state s' at distance $k + 1$ of classical play:



In E_k : the executed verdict is that of the conflict; its execution has emptied A1 and recolored the q-stone black, its stamp kept; the placement has added B3 and A3. In $P(a_0, B3, A3)$ the white A3 has no liberty – A2, A4 and B3 are black – and is gone, a suicide in one order and a capture in the other; write a'_0 for the single outcome. In $P(a_1, B3, A3)$ the white A3 joins A1 and A2, and the component keeps the liberty B2. Every element of $P(a_1, B3, A3)$ occupies A1, empty on the display, and the re-cut of Δ removes them: $\rho(s') = \{a'_0\}$. A3 is empty in a'_0 yet stands on the display. The verdict execution changes only the display. The elements of $P(a_1, B3, A3)$ therefore still stand at the removal round of Δ : they occupy A3 and protect it. They are discarded only at the re-cut, and the recursion stops with the display unchanged. So $I(s') \subsetneq \bigcap_{b \in \rho(s')} \mathbb{I}(b)$, and the radius of k -DSGo is exactly k .

The mechanism, for contrast with $DSGo$: there every stone of the display stands in some branch, the postprocessing keeping the whole decoherence. In $k-DSGo$ the executed verdict empties an intersection of the display alone. The branches still occupying it fail the re-cut; a display stone occupied in those branches alone then stands in no branch.

The proof is now complete. $\square$

Lemma 6.4 (One-stage resolution of commuting turns). *Let s be a semi-classical state of SGo with associated state a , and $\text{diagram}(s) = \text{diagram}(a)$. Let m_0/m_1 be a joint move commuting at a , and let a' be the associated state of $s' = E_{SGo}(s, m_0, m_1)$, per Definition 4.12. Then the resolution of the turn consists of capture reductions at the current turn alone, and $\text{diagram}(s') = \text{diagram}(a')$.*

The proof is elementary but has many moving parts. One can say that the E_{SGo} evolution is fine tuned for the lemma. The reader may recall the example of Section 3.3, which exemplifies the lemma firing – and shows how close the lemma is to failing: without the component clause it simply fails.

Proof. The state a is legal, being reached by classical play. Write A for $\text{diagram}(a)$, n for the current turn, D for the display of s , and D^+ for the display with the turn's stones placed: the stamps of D are below n , the placed stones stamped n . Components, liberties and adjacency refer to D^+ unless a state or a diagram is named.

Step 0 (a commuting joint move places on distinct empty intersections, or passes). In other words, for non pass moves m_0, m_1 to the same intersection i , $a\#m_0\#m_1$ and $a\#m_1\#m_0$ cannot both be defined.

Say $a\#m_0\#m_1$ is defined. Then i is empty in $a\#m_0$: Black's stone is emptied at its own move. A component captured by m_0 is white, trapped in $A + m_0$, and adjacent to i : elsewhere the fill changes no liberties, and a is legal. Its emptied intersections would be liberties of Black's component at the suicide condition (2): Black's stone would stand. So m_0 captures nothing. So Black's component is trapped in $A + m_0$ itself: every black neighbor of i lies in a component of A whose one liberty is i , and no white component of A has i as its one liberty: such a component would be captured. Now play m_1 first. White's placement at i captures those black components (1). The component of White's stone keeps a liberty: an empty neighbor of i , an intersection just freed, or a liberty of a white neighbor other than i . The white stone stands. So i is occupied in $a\#m_1$, and $a\#m_1\#m_0$ is undefined.

In particular, D^+ carries no q-stone, and condition 2 and the q-clauses are vacuous. Call a stone of the turn, and the intersection it occupies, *fresh*; a component of D^+ is *fresh* if it contains a fresh stone, and *old* otherwise. Since a is legal and $\text{diagram}(D) = \text{diagram}(a)$, no component of D is trapped. A component trapped in D^+ therefore contains or neighbors a fresh stone: the turn fills only the fresh intersections. Such a component is trapped on turn n .

The operator reads s from its display, so the hypothesis reads $\text{diagram}(D) = A$. Write A^+ for $\text{diagram}(D^+)$: it is A with the fresh intersections filled in their colors. Components, liberties and trapping of D^+ and of A^+ agree. Write A' for $A\#m_0\#m_1$. The composites $a\#m_0\#m_1$ and $a\#m_1\#m_0$ have the common position normal form a' , and diagram forgets the gradings: $A' = \text{diagram}(a') = A\#m_1\#m_0$.

The placement intersection of m_1 is empty in $A - a\#m_1$ is defined, the turn commuting – and differs from that of m_0 , by Step 0; symmetrically for m_0 . The move m_1 fills only its placement intersection (3). So an intersection empty in $A\#m_0$ and occupied in A is not the placement intersection of m_1 , and is empty in $A\#m_0\#m_1$; and so is the placement intersection of m_0 , when empty in $A\#m_0$. In particular an intersection of $A + m_0$ empty in $A\#m_0$ is empty in A' : it is occupied in A , or is the placement intersection of m_0 . So A' is A^+ with a set of intersections emptied: its stones are stones of A^+ , and no stone recolors.

Step 1 (removal dichotomy). Every component C of A^+ stands entire in A' – its intersections keep their values – or is entirely empty in A' ; an empty one is trapped in A^+ .

For the dichotomy: suppose C stands partial. Being connected, C contains adjacent stones z and w , z occupied and w empty in A' .



Left, the joint move A3/C3. The white pair is one component of $A + m_0$, and is captured whole: adjacent stones share their fate. Right, the joint move B1/A3. The sole liberty of A2 is the placement intersection of m_1 , empty in $A + m_0$: A2 survives m_0 , and stands with the stone of m_1 in A' .

The stone w is a stone of $A + m_0$ empty in $A\#m_0$, or a stone of $(A\#m_0) + m_1$ empty in $A\#m_0\#m_1$; set $Q = A + m_0$ in the first case, $Q = (A\#m_0) + m_1$ in the second. By (1)–(3), w lies in a captured or suicided component R of Q . The component R has no liberties in Q : the suicided component has none even with the captures emptied. The stone z is occupied in A' . The intersections of the captured and suicided components of either move are empty in A' . So z lies in no such component. If z is a stone of Q , it lies in R , adjacent to w and of its color: a contradiction. Otherwise $Q = A + m_0$ and z is the stone of m_1 : its intersection – empty in A , not the placement intersection of m_0 – is empty in Q , adjacent to R : a liberty, a contradiction.

For the trapping: $A + m_0$ and $(A\#m_0) + m_1$ carry at most the stones of A^+ , so a component removed from either has no liberties in A^+ . The stones of an entirely empty component lie in such removed components, and every intersection adjacent to it is adjacent to one of these: it is trapped in A^+ .

Call a component of A^+ *dead* if it is empty in A' , and *live* otherwise.

Step 2 (untrapped components are live). By Step 1 a dead component is trapped in A^+ ; contrapositively, an untrapped component is live.

Step 3 (old trapped components are dead). Let T be old and trapped, white say. The stones of T are stones of A , and T is trapped in A^+ : every liberty of T in A is a fresh intersection. The placement intersection of m_1 is no liberty of T : the white stone there would lie in T 's component, and T is a component of A^+ . The state a is legal: T has a liberty in A . So T has exactly one liberty in A : the placement intersection of the black move m_0 . In $A + m_0$ that liberty is filled: T is an opposite

colored component of it with no liberties, and is captured (1). So the intersections of T are empty in $A\#m_0$, hence in A' : T is dead. A black T is captured by m_1 along $A\#m_1\#m_0$, which is A' as well.

Step 4 (live trapped components are fresh, with a dead donor). Let B be trapped and live. In A' the stones of B stand. A stone of A' is a stone of A^+ , and a stone of A^+ adjacent to B is not of B 's color, B being a component of A^+ : the stones of B are a component of A' . The state a' is legal, so this component has a liberty in A' . That intersection is occupied in A^+ : B is trapped there. Its component in A^+ is dead, by Step 1. Call it a *donor* of B . A donor is adjacent to B , and opposite colored. By Step 3, B is fresh.

The component clause example of Section 3.3 shows the situation: the black component through B3 is trapped and live, and the captured pair is its donor.

Step 5 (no two dead components are adjacent). Suppose dead T and dead Z are adjacent; they are opposite colored, a same colored pair being one component. As for w in Step 1, each stone of a dead component is a stone of $A + m_0$ empty in $A\#m_0$, or a stone of $(A\#m_0) + m_1$ empty in $A\#m_0\#m_1$.

Definition 6.5. We say in the case above that the stone **dies at** m_0 , respectively at m_1 .

Fix adjacent stones x of T and y of Z . First suppose x dies at m_0 and y at m_1 , or vice versa; say the former. Write i for the intersection of x . The intersection i is not the placement intersection j of m_1 : it is occupied in $A + m_0$, and j is not. So i is empty in $(A\#m_0) + m_1$, and the component of y in the latter has i as a liberty. But y dies at m_1 : its component is captured or suicided, with no liberties there, as in Step 1. A contradiction. So x and y die at one move; write m for it.

One of x and y has the color of m , the other the opposite; say y does. The stone y is emptied, and its component at m is not captured: captured components are opposite colored (1). So that component is suicided: it is the component of the placed stone (2). And x , opposite colored, is captured: an emptied stone lies in a captured or suicided component (3), and the suicided one has the color of m . The suicide condition (2) reads the placed diagram with the captures emptied. There the placed stone's component is adjacent to the emptied intersection of x , a liberty, and does not suicide: a contradiction.

Step 6 (the reduction removes only dead components). Write D_k for $\widetilde{Cap}_n^k(D^+)$. We show by induction on k : every stone of D^+ empty in D_k lies in a dead component, and every dead component is empty in D_k entire, or stands in D_k entire. For $k = 0$ there is nothing to show. Suppose the claim holds at k . By Step 5, removals of dead components never free a dead component: every dead component standing in D_k is trapped in D_k . It also stays trapped on turn n . A dead fresh component contains a stone of the turn. A dead old component is adjacent, by Step 3, to the opposite colored stone of the turn at its one liberty of A . That stone's component is live by Step 5, and stands in D_k , by the claim at k . Now let $\widetilde{Cap}_n$ capture C in D_k , and suppose a stone of C were live: its component B of A^+ stands in D_k , by the claim at k , and $C = B$: B is connected in D_k , and C in A^+ . Trapped in D_k , B is trapped in A^+ , whose empty intersections are empty in D_k . By Step 4, B is fresh, with a dead donor T adjacent. B fresh blocks condition 3b: a stone of B is of the turn. If T is empty in D_k , its intersections are liberties of C : not trapped. If T stands in D_k , entire by the claim at k , it is still trapped on turn n , and condition 3a fails for C . Either way C is not captured: a contradiction. So every stone of C

is dead. The dead component of one of them stands entire in D_k ; it is connected in D_k , so it lies in C ; and C , connected in A^+ , lies in it: C is that component. The removals stay entire, and the claim holds at $k + 1$.

Step 7 (progress, and at most two removing applications). Consider the first application, D_0 to D_1 . Let T be a dead old component, white say. Its adjacent opposite colored stones trapped in D_0 lie in live components: dead neighbors do not exist, by Step 5. Live trapped components are fresh, by Step 4. So condition **3b** holds for T : no stone of T is of the turn, and every adjacent trapped component contains a stone of the turn that is not a q-stone. Every dead old component is captured on the first application.

Now let T be a dead fresh component, through m_1 say. Its possible blocker for condition **3a** is an adjacent trapped opposite colored component; by Step 5 it is live, by Step 4 fresh: the component Z of the stone of m_0 . Suppose Z is adjacent to T and trapped in D_0 . Some dead old white component is adjacent to Z . Suppose not, and work along $A \# m_0 \# m_1$. A white captured by m_0 is dead and old, its stones being stones of A . It is not of T : an old white part of T is a component of A , and T is connected: the part is adjacent to the stone of m_1 , and its placement intersection, empty in $A + m_0$, is a liberty of the part. So none is adjacent to Z . Then either Z has no liberties at the suicide condition of m_0 (**2**) and suicides, or its one liberty is the placement intersection of m_1 , filled by m_1 , whose captures then remove Z (**1**). In both cases Z is empty in A' , hence dead: a contradiction. The adjacent dead old white component is captured on the first application, freeing Z : in D_1 the component Z is no longer trapped, condition **3a** holds for T , and T is captured.

So every dead component is empty in D_2 : at most two applications remove stones, and $Cap_n(D^+)$ empties exactly the dead components.

Step 8 (later stages are quiet, and conclusion). The removals at turn n only emptied intersections. The stones standing are the live components: by Steps 1–4, exactly the stones of A' . The state a' is legal: no component of A' is trapped. So Cap_t removes nothing for $t < n$, and the recursion stops.

No recoloring occurred, there being no q-stone. So

$$\text{diagram}(OR(D^+)) = \text{diagram}(a').$$

And $OR(D^+)$ is the display of s' : $\text{diagram}(s') = \text{diagram}(a')$. If a move is a pass, the same argument runs with one fresh intersection. $\square$

Remark 6.6. *Step 7 shows the repetition needs at most two removing applications on any board: there are at most two fresh components, and the second application serves only a dead fresh component shielded by the other. Step 8 shows the later stages are quiet: at a commuting turn the recursion depth of Section 2.1 is at most one, as the statistic there suggests.*

6.2. Chess and faithfulness. We now come to the main reason for studying faithfulness: it gives rise to a qualitative mathematical difference between chess and Go.

Theorem 6.7. *$\widetilde{\text{chess}}$ has no simple simultaneization that is faithful within distance one of classical play: equivalently, none of positive faithfulness radius. The same holds for its minimal quotient. In particular, the universal simultaneization $\text{Sim}(\widetilde{\text{chess}})$, faithful within every distance (Lemma 4.16), is not simple.*

8	.	n	b	q	k	b	n	r
7	.	p	p	p	p	p	p	p
6	.	r	.	.	.	.	.	.
5	p	.	.	.	.	.	.	.
4	P	.	.	.	.	.	.	.
3	.	R	.	.	.	.	.	.
2	.	P	P	P	P	P	P	P
1	.	N	B	Q	K	B	N	R
	a	b	c	d	e	f	g	h

FIGURE 1. The state a^* , reached by the mirror pairs $a4/a5$; $Ra3/Ra6$; $Rb3/Rb6$. White pieces are in capitals. The conflicting simultaneous moves are the rook moves $m_0 : b3 \rightarrow b5$ and $m_1 = \mathcal{A}(m_0) : b6 \rightarrow b4$.

Proof. First a technical lemma.

Lemma 6.8. *Let G_1 be a simultaneization of G_0 , and suppose that $s \in B(G_1)$ is obtained from the initial state by a game history consisting of mirror pairs, that is, of simultaneous pairs of the form $(x_i, \mathcal{A}(x_i))$. Then $\mathcal{R}(s) = s$, and consequently $\mathcal{A}(I(s)) = I(s)$.*

Proof. The mirror of the pair $(x, \mathcal{A}(x))$ is the pair $(\mathcal{A}(\mathcal{A}(x)), \mathcal{A}(x)) = (x, \mathcal{A}(x))$ itself. So applying the symmetry property of Definition 4.2, and the order invariance of E (the last property of Definition 4.1), along the game history, together with $\mathcal{R}(q_0) = q_0$, we get $\mathcal{R}(s) = s$. Then for any m , $E_{G_1}(s, m)$ is defined if and only if $E_{G_1}(\mathcal{R}(s), \mathcal{A}(m)) = E_{G_1}(s, \mathcal{A}(m))$ is defined, giving $\mathcal{A}(I(s)) = I(s)$. $\square$

Now return to the proof of the theorem. Let G_1 be any simultaneization of $\widetilde{\text{chess}}$ faithful within distance one of classical play, with P_0 the White player, and consider the game history of G_1 consisting of the mirror pairs

$$a4/a5; \quad Ra3/Ra6; \quad Rb3/Rb6,$$

in standard chess algebraic notation. The moves of each pair do not interact: both composites $a\#m_0\#m_1$ and $a\#m_1\#m_0$ are defined, with the same position normal form, so each turn commutes (Definition 4.12) and each state along this history is semi-classical. By the semi-classicality condition of Definition 4.13 together with faithfulness at the semi-classical states, applied inductively along the history, the history is legal in G_1 and for the resulting state $s^* \in B(G_1)$ we have $\rho(s^*) = \{a^*\}$, with a^* the classical state of Figure 1. Note that this pins down $\rho(s^*)$ for every simultaneization G_1 , whatever its branch semantics.

Now let m_0 be the White rook move $b3 \rightarrow b5$, and $m_1 = \mathcal{A}(m_0)$ the Black rook move $b6 \rightarrow b4$. Each order of these moves blocks the other: after $b3 \rightarrow b5$ the Black rook's path is blocked at $b5$, while after $b6 \rightarrow b4$ the White rook's path is blocked at $b4$. So in $\text{Sim}(\widetilde{\text{chess}})$ this pair resolves by the lowest priority case of the serialization map P , and

$$P(a^*, m_0, m_1) = \{a_0, a_1\},$$

where in a_0 the White rook stands on $b5$ with the Black rook on $b6$, and a_1 is the mirror image of a_0 . By faithfulness at the semi-classical s^* , the pair (m_0, m_1) ,

being available in $\text{Sim}(\widetilde{\text{chess}})$ at $\{a^*\}$, is available at s^* ; so $s_1^* = E_{G_1}(s^*, m_0, m_1)$ is defined, and by condition 3 together with faithfulness at s_1^* – a state one turn from $s^* - \rho(s_1^*)$ is a nonempty subset of $\{a_0, a_1\}$. Moreover s_1^* is not final: the faithfulness equation holds at s_1^* ; neither a_0 nor a_1 is final in $\widetilde{\text{chess}}$, so the right side contains the pass, and the left side is nonempty.

Suppose that $\rho(s_1^*)$ is a singleton, say $\{a_0\}$. The history of s_1^* consists of mirror pairs, so by Lemma 6.8 $I(s_1^*)$ is $\mathcal{A}$ invariant. But by faithfulness at s_1^* , $I(s_1^*) = \mathbb{I}(a_0)$, which is not $\mathcal{A}$ invariant: in a_0 White has rook moves from $b5$, while Black has no rook moves from $b4$. The case of $\{a_1\}$ is mirrored. Thus a singleton value is forbidden by the symmetry of G_1 , and $\rho(s_1^*) = \{a_0, a_1\}$.

By faithfulness at s_1^* , $I(s_1^*) = \mathbb{I}(a_0) \cap \mathbb{I}(a_1)$, which is computed directly: for each player it consists of the pass move, the moves of the c through h pawns, and the four knight moves. There are no queen side rook moves, as these rooks stand on different squares in a_0 and a_1 , and no a or b pawn moves.

Now suppose that G_1 is simple, so that there is a legally obtainable state c of $\widetilde{\text{chess}}$ with $\mathbb{I}(c)$ equal to this set. The interface $\mathbb{I}(c)$ forces every piece of c other than the two queen side rooks, the two a pawns and the two b pawns to stand on its starting square. One now checks the finitely many placements, or absences by capture, of the six remaining pieces, and each fails. For example, if the White b pawn is at $b2$, then $b3$ must be occupied, necessarily by a queen side rook or the Black b pawn, giving respectively a legal rook move from $b3$ and the legal capture $b3 \times c2$, neither of which is in the set; while if the White b pawn is absent, having been captured, and $b2$ is unoccupied, then the $c1$ bishop has the legal move to $b2$, again not in the set. So we have a contradiction, and G_1 is not simple.

For the minimal quotient the proof applies as it stands. It reads the definedness of composites, interfaces, finality and legal obtainability, which π preserves, Lemma 4.20(4), and the inequality $[a_0] \neq [a_1]$, which holds as $\mathbb{I}(a_0) \neq \mathbb{I}(a_1)$. The witness of the last step is a class $[c]$, with $\mathbb{I}([c]) = \mathbb{I}(c)$; the check is the same. $\square$

6.3. Zero in the spectrum. Let G be a sequential game. Its *truncated universal simultaneization* $\text{Sim}_{sc}(G)$ is $\text{Sim}(G)$ with every state that is not semi-classical declared final: it evolves as $\text{Sim}(G)$ at the semi-classical states and nowhere else, the final states of $\text{Sim}(G)$ are scored as there, and the new final states are drawn. Play proceeds classically while the turns commute; the first turn that does not ends the game.

Theorem 6.9. *Let G be a sequential game.*

- (1) $\text{Sim}_{sc}(G)$ is a simple simultaneization of G , faithful within distance zero.
- (2) Suppose that at some semi-classical state $\{a\}$ of $\text{Sim}(G)$ a joint move m_0/m_1 is available with $P(a, m_0, m_1)$ having two elements, neither final. Then $\text{Sim}_{sc}(G)$ has faithfulness radius zero, and so 0 is in the raw radius spectrum of G . If the core states of the two elements are not bisimilar, 0 is in the radius spectrum of G .
- (3) Suppose instead that every joint move available at a semi-classical state of $\text{Sim}(G)$ commutes at its associated state. Then the raw radius spectrum of G , and its radius spectrum, are $\{\infty\}$.

Proof. (1) Write $G_1 = \text{Sim}_{sc}(G)$. Its non final states are the non final semi-classical states of $\text{Sim}(G)$: singletons $\{a\}$ with a non final, by Lemma 4.16. The conditions

of Definition 4.1 are inherited from $\text{Sim}(G)$, the evolution being cut at the new final states; at a non final $\{a\}$ the pass is available, a being non final. Let $\tilde{\mathcal{A}}, \tilde{\mathcal{R}}$ be the operators of Lemma 4.10. By (4.11), and as $\mathcal{R}$ commutes with the position normal form up to the turn grading, $\tilde{\mathcal{R}}$ carries commuting turns to commuting turns. So it preserves the semi-classical states, the states of G_1 and the final states of $\text{Sim}(G)$, hence the new final states, and G_1 is symmetric. Condition 2 holds by construction. Let ρ be the inclusion of $B(G_1)$ in $2^{N(S(G))}$. Condition 3 holds, E_{G_1} being a restriction of $E_{\text{Sim}(G)}$, and $\rho(q_0) = \{q_0\}$. The semi-classical states of G_1 are those of $\text{Sim}(G)$, with the same associated states, the two evolutions agreeing there: condition 4. So G_1 is a simultaneization.

Faithfulness at a semi-classical $\{a\}$: $\rho(\{a\}) = \{a\}$ is nonempty, and $I(\{a\}) = \mathbb{I}(a)$: both sides are empty when a is final, $\{a\}$ being final in G_1 exactly when in $\text{Sim}(G)$; otherwise the equality is that of $\text{Sim}(G)$, Lemma 4.16. So G_1 is faithful within distance zero.

Simplicity: let $\{a\}$ be non final. Then $I(\{a\}) = \mathbb{I}(a)$. The interface of a state of pass grading $j < 2$ depends on its core state alone, by property 2 of Definition 4.3 and the formula for $\#$. The core state of a is legally obtainable, a being a position normal form, and some legally obtainable state a'' of pass grading $j < 2$ lies over it: a legally obtainable $(x, i, 2)$ follows the legally obtainable $(x, i + 1, 1)$. So $\mathbb{I}(a'') = I(\{a\})$.

(2) The joint move is available at $\{a\}$ in G_1 and leads to $\mathfrak{s}' = P(a, m_0, m_1)$, a state one turn from a semi-classical state. It has two elements, so it is not semi-classical (Lemma 4.16), and is final in G_1 : $I(\mathfrak{s}') = \emptyset$. No branch of $\rho(\mathfrak{s}') = \mathfrak{s}'$ is final, so the pass lies in every $\mathbb{I}(a)$, $a \in \mathfrak{s}'$. The faithfulness equation fails at $\mathfrak{s}'$: G_1 is not faithful within distance one. With (1), its radius is zero. For the last claim apply the above to $G/\sim$: by Lemma 4.20(4), $\{[a]\}$ is a semi-classical state of $\text{Sim}(G/\sim)$ at which m_0/m_1 is available, with $P([a], m_0, m_1)$ the image of $P(a, m_0, m_1)$: two elements, the core states of the two being not bisimilar, neither final.

(3) Let G_1 be a simultaneization of G faithful within distance zero, s semi-classical with associated state a , and $s' = E_{G_1}(s, m_0, m_1)$ defined. By condition 3 the joint move is then available in $\text{Sim}(G)$ at $\rho(s)$, which is $\{a\}$ by condition 4. By hypothesis the joint move commutes at a , so s' is semi-classical. By recursion every state is semi-classical, and hence its radius is ∞ . Applied to $\text{Sim}(G)$ itself, faithful within distance zero by Lemma 4.16, this shows that every state of $\text{Sim}(G)$ is semi-classical. So $\text{Sim}_{sc}(G)$ is $\text{Sim}(G)$: simple by (1), of radius ∞ by Lemma 4.16. The raw radius spectrum is $\{\infty\}$. Every legally obtainable state of $\text{Sim}(G/\sim)$ is the image of a semi-classical $\{a\}$, Lemma 4.20(4), and a joint move available there is available at $\{a\}$, where it commutes at a , hence at $[a]$. So $G/\sim$ satisfies the hypothesis, and the radius spectrum of G is $\{\infty\}$. $\square$

Example 6.10. *Let each player own a board of n cells; a non pass move fills an empty cell of the mover's board, the pass is always defined, and the game ends by a double pass. At $a = ((B_0, B_1), 0, 0)$ both composites of a joint move m_0/m_1 have the core $(B_0 + m_0, B_1 + m_1)$, and both composites of $m_0/0$ the core $(B_0 + m_0, B_1)$: every joint move commutes. By Theorem 6.9(3) its raw radius spectrum and its radius spectrum are $\{\infty\}$; neither contains 0.*

For $\widetilde{\text{chess}}$ the hypothesis of Theorem 6.9(2) holds at the state a^* of Figure 1, semi-classical by the proof of Theorem 6.7, with the rook pair: $P(a^*, m_0, m_1) =$

$\{a_0, a_1\}$, neither final, and of non bisimilar core states: $\mathbb{I}(a_0) \neq \mathbb{I}(a_1)$, as in that proof, while bisimilar states have the same interface, Lemma 4.20(4). With Theorem 6.7, the raw radius spectrum and the radius spectrum of $\widetilde{\text{chess}}$ are $\{0\}$, while those of Go are all of $\mathbb{N} \sqcup \{\infty\}$, by Theorem 6.3.

Question 6.11. *Which subsets of $\mathbb{N} \sqcup \{\infty\}$ arise as radius spectra? $\widetilde{\text{chess}}$ and Go realize $\{0\}$ and the whole of $\mathbb{N} \sqcup \{\infty\}$, and a game whose available turns all commute realizes $\{\infty\}$, Theorem 6.9. We do not know whether the spectrum of a game with a conflict of non bisimilar outcomes, as in Theorem 6.9(2), must be an initial segment. Nor do we know a game of finite positive absolute radius – a spectrum with finite positive supremum – but it is hard to imagine an obstruction.*

How does the radius spectrum sit among known invariants of games? The classical invariants are functionals of the scoring. The Conway value, the temperature, the outcome class: each is computed from the terminal values, by backward induction [4, 2]. The same holds for the invariants obtained by perturbing the turn mechanism. Under auction play, the Richman value is a threshold for winning [21, 5]. Under random turn order, the value is a winning probability [27]. The complexity class of a game family is the cost of deciding the winner; notably it does not draw our line, as generalized chess and Go with a simple Ko rule are both EXPTIME complete [10, 31]. The radius spectrum is of the same species – an invariant born of perturbing the turn mechanism – but it alone is blind to the scoring. It reads the dynamics alone, up to bisimulation (Lemma 4.20). Purely dynamical equivalences of automata are the subject of concurrency theory: bisimulation, and the trace theory of commuting actions [23, 22]. A commuting turn is a diamond in that language, and $\text{Sim}(G_0)$ resembles the passage from interleaving semantics to true concurrency. The raw radius spectrum reads the diamonds of the state space, which a bisimulation may collapse, Remark 6.12; the radius spectrum reads those of the minimal automaton. We do not know a numerical invariant of trace theory in the role of the radius.

Remark 6.12. *The raw radius spectrum is not an invariant of bisimulation. Let G' be the game of Example 6.10 and G the same game with a record added to the core state: the color of the last stone placed, empty initially. Forgetting the record is a bisimulation between G and G' , Definition 4.19: no play distinguishes the two games. In G a joint move of two placements does not commute, its two orders differing in the record. So 0 lies in the raw radius spectrum of G , by Theorem 6.9(2) at the initial state, and not in that of G' . The record is objective information, the order of play written into the state, and the radius measures exactly such information. The radius spectrum forgets it: both games have radius spectrum $\{\infty\}$, by Theorem 6.9(3) for G' and Lemma 4.20(2).*

Lemma 6.13. *Go is minimal: distinct legal diagrams are not bisimilar.*

Proof. The core states of Go are the legally obtainable diagrams, and these are legal, by induction along the play: when every component of B has a liberty, so has every component of $B \# m$, by (1)–(3). Let $D \neq D'$ be legal diagrams and $\sim$ a bisimulation on Go with $D \sim D'$. A bisimulation preserves the legality of move sequences, by induction on the length: a move sequence is legal from $(D, 0, 0)$ exactly when it is legal from $(D', 0, 0)$. We produce a sequence legal from one state only.

If some intersection is empty in one diagram and occupied in the other, Black’s move to it is the sequence. So let the occupancies agree, and, interchanging D and D' if necessary, let some intersection i carry a black stone in D and a white one in D' . Let K be the black component of i in D and j a liberty of K , which exists, D being legal.

Play, from both states, the sequence of moves described next, and stop as soon as the occupancies of the two evolved diagrams differ. While they agree, every placement of the sequence is legal from both states, a placement being legal exactly when its intersection is empty, and every pass is legal, no state being final: no two passes of the sequence are consecutive. Once the occupancies differ, a move to an intersection empty in one diagram only is legal from one state only, and we are done.

Write K for the black component of i in the evolved D . A step chooses a liberty $l \neq j$ of K and fills it: by White, if White’s stone at l would stand in the evolved D , and otherwise by Black. The other player passes first when it is their turn. Black’s stone at l stands: it joins K , which has the liberty j . So K only grows, j stays its liberty, j is never filled, and K is never captured. Each Black placement adds a stone to K , so there are finitely many. Between two of them K is fixed, and each White placement fills one of its liberties and frees none, the black stones it removes lying in components other than K , not adjacent to K . So after finitely many steps j is the sole liberty of K .

White now plays j , Black passing first if it is Black’s turn. In the evolved D the component K is captured, by (1), and i is empty. In the evolved D' the stone at i is still white, the occupancies having agreed, and a White placement removes no white stone but its own. Black’s move to i is then legal from one state only. $\square$

7. INFORMED PLAY AND CONCLUDING REMARKS

The focus of this section is our informed play experiments: their methodology and partial results, for both games. All figures below are for the 5×5 board, and may be compared directly with the random play columns of the table of Section 2.1. The restriction to 5×5 is due to computational cost: for *SGO*, estimating the equilibrium with our very blunt methods is expensive even on 5×5 .

7.1. Equilibrium guided play. Our equilibrium guided play is computed as follows: at each turn both players are offered a random menu of five moves; the resulting 5×5 matrix of joint outcomes is filled by averaging the final margins of short uniformly random continuations;⁸ and each player samples their move from an optimal mixed strategy of this zero sum matrix game. Three technical findings. First, at about 74% of the turns the estimated one turn game has *no pure saddle point*, so the players must randomize – consistent with Section 5.1, where mixing is proved necessary already on the 3×3 board (re-estimating the matrices at fivefold rollout depth, on fixed states and menus, lowers the mixing fraction from 74% to 54% on a control sample; genuine mixing remains). Second, the within turn value spread is large, about 16 points of the twenty-five: the joint move choice of a single turn swings most of the board. Third, the summary table of the next subsection

⁸Rollouts use the standard Monte Carlo playout convention: uniformly random moves, except that a player does not fill their own single point eyes and does not play suicide, passing when no other move is available.

shows that equilibrium guided play leaves the creation of entanglement essentially unchanged relative to the random play columns of the table of Section 2.1 (3.2 q-stones created per game against 3.3): the entanglement is a stable feature of the game, of order one under random and informed play alike.

7.2. Versus informed play in classical Go. For the classical game we use greedy rollout guided play: at each move the player evaluates up to six candidate moves, and the pass, by twelve random rollouts each (the convention of the preceding footnote), and plays the best. The summary of informed self play, for both games, in the format of the table of Section 2.1 (162 and 165 games; unbounded play in both cases; these runs predate the depth statistic and report branch counts instead):

	<i>SGo</i>	classical <i>Go</i>
mean margin	11.7	11.4
draws	5.6%	6.7%
games ending on their own	100%	100%
mean length in turns	21.2	17.8
q-stones created per game	3.2	–
q-stones on the final display	2.3	–
peak number of branches	4.2	–
final number of branches	3.2	–

For *SGo* the table is close to its random play columns: the margin eases (11.7 against 13.6), the draw rate and game length barely move, and the entanglement is unchanged in creation and slightly more resolved by the end. For the classical game the change is termination: informed play supplies the endings that random play never reaches – every game now ends on its own, in 17.8 turns on average – while *SGo* terminates by rule either way.

At this shallow search depth no measurable first mover advantage emerges in *Go* (signed mean margins statistically zero). Hence we are not correcting by any Komi, which is also why draws appear for *Go* in the table above. We note that *Go* is solved on the 5×5 board – a win for Black by 25 points under standard rule formalizations [39] – a reminder that these are partial, shallow search depth results.

Whether the above observations persist for equilibrium guided play on large boards, along with the sharpness observed in Section 2.1, we leave to future study.

7.3. An Elo ladder. Sharp dynamics alone do not distinguish a deep game from a coin flip. A standard proxy for depth is the length of the improvement ladder: how much playing strength a fixed family of agents can climb before it saturates [20]; strength against computational budget is likewise standard in the study of game-playing programs [17]. We ran one ladder for both games, on the 5×5 board, with the same architecture: Monte Carlo agents indexed by a rollout budget, choosing from a random menu of five moves by random playouts, the opponent modeled as random. Adjacent budgets played 100 games each, colors alternating; the final pair played 60. The Elo gap of a pair is $400 \log_{10}(s/(1-s))$, with s the stronger agent’s mean score.

budgets	<i>SGo</i>	<i>Go</i>
0 v. 16	+308	+173
16 v. 128	+330	+89
128 v. 1024	+215	+63
1024 v. 2048	+52	−45
2048 v. 4096	0	−
ladder	≈ 900	≈ 300

Both ladders saturate. That of *Go* does so by budget 1024: its last rung is negative, within noise of zero. Direct matches between non adjacent rungs agree with the summed gaps, in both games: the ladders hide no cycles. A control with komi one half, which eliminates draws, leaves the ladder of *Go* essentially unchanged: the disparity is not an artifact of the draw channel.

These figures must be properly interpreted. They are relative to this agent family, and they lower bound the depth rather than measure it. Still, on the same board and by the same yardstick, the ladder of *SGo* is three times the height, and the scale is exponential in nature.

7.4. Lazy evaluation. Finally, we note that the pattern of resolution in *SGo* – branchings accumulate silently until objective information forces their evaluation – is also familiar in computer science, notably in lazy evaluation [12, 15], where a deferred computation is forced exactly when its value is demanded.

REFERENCES

- [1] R. ALUR, T. A. HENZINGER, AND O. KUPFERMAN, *Alternating-time temporal logic*, Journal of the ACM, 49 (2002), pp. 672–713. 1.2, 4
- [2] E. R. BERLEKAMP AND D. WOLFE, *Mathematical Go: Chilling Gets the Last Point*, AK Peters, Natick, MA, 1994. 1.1, 6.3
- [3] A. CINCOTTI AND H. IIDA, *The game of synchronized cutcake*, in IEEE Symposium on Computational Intelligence and Games (CIG 2007), IEEE, 2007, pp. 374–379. 1.1
- [4] J. H. CONWAY, *On Numbers and Games*, Academic Press, 1976. 1.1, 4.1, 6.3
- [5] M. DEVELIN AND S. PAYNE, *Discrete bidding games*, Electronic Journal of Combinatorics, 17 (2010), p. R85. 6.3
- [6] L. DOYEN AND J.-F. RASKIN, *Games with imperfect information: theory and algorithms*, in Lectures in Game Theory for Computer Scientists, Cambridge University Press, 2011. 4.9
- [7] J. EISERT, M. WILKENS, AND M. LEWENSTEIN, *Quantum games and quantum strategies*, Physical Review Letters, 83 (1999), pp. 3077–3080. 1.1
- [8] C. A. ELLIS AND S. J. GIBBS, *Concurrency control in groupware systems*, in Proceedings of the 1989 ACM SIGMOD International Conference on Management of Data, ACM Press, 1989, pp. 399–407. 1.2
- [9] A. M. FINK, *Equilibrium in a stochastic n-person game*, Journal of Science of the Hiroshima University, Series A-I, 28 (1964), pp. 89–93. 1.2
- [10] A. S. FRAENKEL AND D. LICHTENSTEIN, *Computing a perfect strategy for $n \times n$ chess requires time exponential in n* , Journal of Combinatorial Theory, Series A, 31 (1981), pp. 199–214. 6.3
- [11] D. GALE, H. W. KUHN, AND A. W. TUCKER, *On symmetric games*, in Contributions to the Theory of Games, Vol. I, vol. 24 of Annals of Mathematics Studies, Princeton University Press, 1950. 5.1
- [12] P. HENDERSON AND J. H. MORRIS, JR., *A lazy evaluator*, in Proceedings of the 3rd ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages (POPL ’76), ACM, 1976, pp. 95–103. 7.4
- [13] M. HUGGAN, R. J. NOWAKOWSKI, AND P. OTTAWAY, *Simultaneous combinatorial game theory*. arXiv:1810.02870, 2018. 1.1, 4

- [14] M. A. HUGGAN, *Studies in Alternating and Simultaneous Combinatorial Game Theory*, ph.d. thesis, Dalhousie University, Halifax, Nova Scotia, Canada, 2019. 1.1
- [15] J. HUGHES, *Why functional programming matters*, The Computer Journal, 32 (1989), pp. 98–107. 7.4
- [16] D. R. JEFFERSON, *Virtual time*, ACM Transactions on Programming Languages and Systems, 7 (1985), pp. 404–425. 1.2
- [17] A. L. JONES, *Scaling scaling laws with board games*. arXiv:2104.03113, 2021. 7.3
- [18] H. T. KUNG AND J. T. ROBINSON, *On optimistic methods for concurrency control*, ACM Transactions on Database Systems, 6 (1981), pp. 213–226. 1.2
- [19] L. LAMPORT, *Time, clocks, and the ordering of events in a distributed system*, Communications of the ACM, 21 (1978), pp. 558–565. 1.2
- [20] F. LANTZ, A. ISAKSEN, A. JAFFE, A. NEALEN, AND J. TOGELIUS, *Depth in strategic games*, in Workshops of the Thirty-First AAAI Conference on Artificial Intelligence: What’s Next for AI in Games?, 2017. 7.3
- [21] A. J. LAZARUS, D. E. LOEB, J. G. PROPP, W. R. STROMQUIST, AND D. H. ULLMAN, *Combinatorial games under auction play*, Games and Economic Behavior, 27 (1999), pp. 229–264. 6.3
- [22] A. MAZURKIEWICZ, *Trace theory*, in Petri Nets: Applications and Relationships to Other Models of Concurrency, Advances in Petri Nets 1986, vol. 255 of Lecture Notes in Computer Science, Springer, 1987. 6.3
- [23] R. MILNER, *Communication and Concurrency*, Prentice Hall, 1989. 6.3
- [24] J. F. J. NASH, *Equilibrium points in n -person games*, Proc. Natl. Acad. Sci. USA, 36 (1950), pp. 48–49. 1.1, 1.2, 5.1
- [25] M. J. OSBORNE AND A. RUBINSTEIN, *A Course in Game Theory*, MIT Press, 1994. 5.1
- [26] R. PENROSE, *The Emperor’s New Mind*, Oxford University Press, 1989. 1.1
- [27] Y. PERES, O. SCHRAMM, S. SHEFFIELD, AND D. B. WILSON, *Random-turn hex and other selection games*, American Mathematical Monthly, 114 (2007), pp. 373–387. 6.3
- [28] M. O. RABIN AND D. SCOTT, *Finite automata and their decision problems*, IBM Journal of Research and Development, 3 (1959), pp. 114–125. 4.9
- [29] A. RANCHIN, *Quantum Go*. arXiv:1603.04751, 2016. 1.1
- [30] J. H. REIF, *The complexity of two-player games of incomplete information*, Journal of Computer and System Sciences, 29 (1984), pp. 274–301. 4.9
- [31] J. M. ROBSON, *The complexity of go*, in Information Processing 83, Proceedings of the IFIP 9th World Computer Congress, 1983, pp. 413–417. 1.1, 6.3
- [32] U. SCHWALBE AND P. WALKER, *Zermelo and the early history of game theory*, Games and Economic Behavior, 34 (2001), pp. 123–137. 5.1
- [33] M. SHAPIRO, N. PREGUIÇA, C. BAQUERO, AND M. ZAWIRSKI, *Conflict-free replicated data types*, in Stabilization, Safety, and Security of Distributed Systems (SSS 2011), vol. 6976 of Lecture Notes in Computer Science, Springer, 2011, pp. 386–400. 1.2
- [34] L. S. SHAPLEY, *Stochastic games*, Proceedings of the National Academy of Sciences, 39 (1953), pp. 1095–1100. 1.2, 4
- [35] Y. SHOHAM AND K. LEYTON-BROWN, *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*, Cambridge University Press, 2009. 1.2
- [36] M. THIELSCHER, *The general game playing description language is universal*, in Proceedings of the 22nd International Joint Conference on Artificial Intelligence (IJCAI 2011), AAAI Press, 2011. 4
- [37] J. TROMP AND G. FARNEBÄCK, *Combinatorics of Go*, in Computers and Games (CG 2006), vol. 4630 of Lecture Notes in Computer Science, Springer, 2007, pp. 84–99. 6.1
- [38] J. TROMP AND B. TAYLOR, *The logical rules of Go*. <https://tromp.github.io/go.html>, 1995. 6.1
- [39] E. C. D. VAN DER WERF AND M. H. M. WINANDS, *Solving Go for rectangular boards*, ICGA Journal, 30 (2009), pp. 77–88. 7.2
- [40] J. VON NEUMANN AND O. MORGENSTERN, *Theory of Games and Economic Behavior*, Princeton University Press, 1944. 1.2